

LiNC

Operator Manual

LinuxCNC HMI • operator panel • CNC keyboard • toolpath • MDI • touch probe • tool table

End-user documentation for the LiNC controller interface and the panel that drives it.

Edition 1.0

Safety and Liability Disclaimer

Read this before you use LiNC on a machine.

LiNC controls CNC machines and other potentially dangerous equipment. By using it you acknowledge and agree to the following.

> ⚠ **SAFETY**
>
> A CNC machine moves rapidly and with great force. It can crush and trap, it throws broken tools and workpiece fragments, a spindle can start a fire, its electrical installation is hazardous, and cutting produces noise and dust. Do not operate CNC equipment unless you are trained and qualified to do so.

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2. **Testing and validation**: proving a program in a safe place before it runs on production equipment.
3. **Supervision**: never leaving a CNC machine running unattended.
4. **Compliance**: the laws, regulations and standards that apply where the machine is installed.
5. **Professional review**: having the installation reviewed by qualified engineers or safety professionals before it is used in a critical application.
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The emergency stop, the guards, the door interlocks, the limit switches and the machine's own safety circuits are hardware. They must stay functional and independent of this software, and they must never be bypassed. LiNC cannot see everything the machine does, cannot prevent every programming error, and its interlocks (including the network interlock) are convenience features, not safety functions.

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The first start shows this notice and asks for acceptance. The answer is recorded, with the version of the notice, in the profile's preferences file (`~/.config/linc/<profile>.preferences.toml`). The command-line option `--accept-safety-notice` records the same acceptance without the dialog, for provisioning and kiosks. When the notice changes, its version changes with it and the dialog asks once more.

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> **NOTE**
>
> This notice is not legal advice. If you deploy LiNC commercially, in a regulated environment, or where its failure could injure someone, have the terms reviewed before commissioning.

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1 Introduction

LiNC is the operator interface of a LinuxCNC machine. It shows what the machine is doing, it loads and edits G-code, and it holds the tool, offset and probing pages. It runs on the control PC as a single program: there is no helper process, no Python layer and no web service between the screen and the controller.

LiNC is operated from three surfaces, and none of them is a touch screen:

1. The **operator panel** – power, emergency stop, mode selector, override knobs, jog keys, cycle keys and the machine function keys. This is where the machine is controlled.
2. The **CNC keyboard** – navigation, G-code entry, MDI, tool and offset values, file handling. Every action on the screen has a key.
3. The **screen** – readouts, the toolpath, the program, the tables and the diagnostics. It reports; it does not control motion.

The layout and the key handling follow the conventions of an industrial machine controller, so an operator who knows a conventional control can work without learning a new interaction model. A mouse is not required, and a touch screen is not required.

The panel and the interface can also run on one computer without a machine cabinet: the *simulated panel* is a second window that carries the same keys, knobs, lamps and displays, and publishes the same interface to the controller. On a machine without a physical panel it stands in for one; a touch screen or a mouse then works as a replacement for the panel keys. It is a replacement for a panel, not a second way to drive the machine.

Because the machine controls stay off the screen, the screen is free for machine state. The digital readout, the toolpath, the program listing, the modal codes and the servo load are visible at the same time, which is what supervising a running job needs.

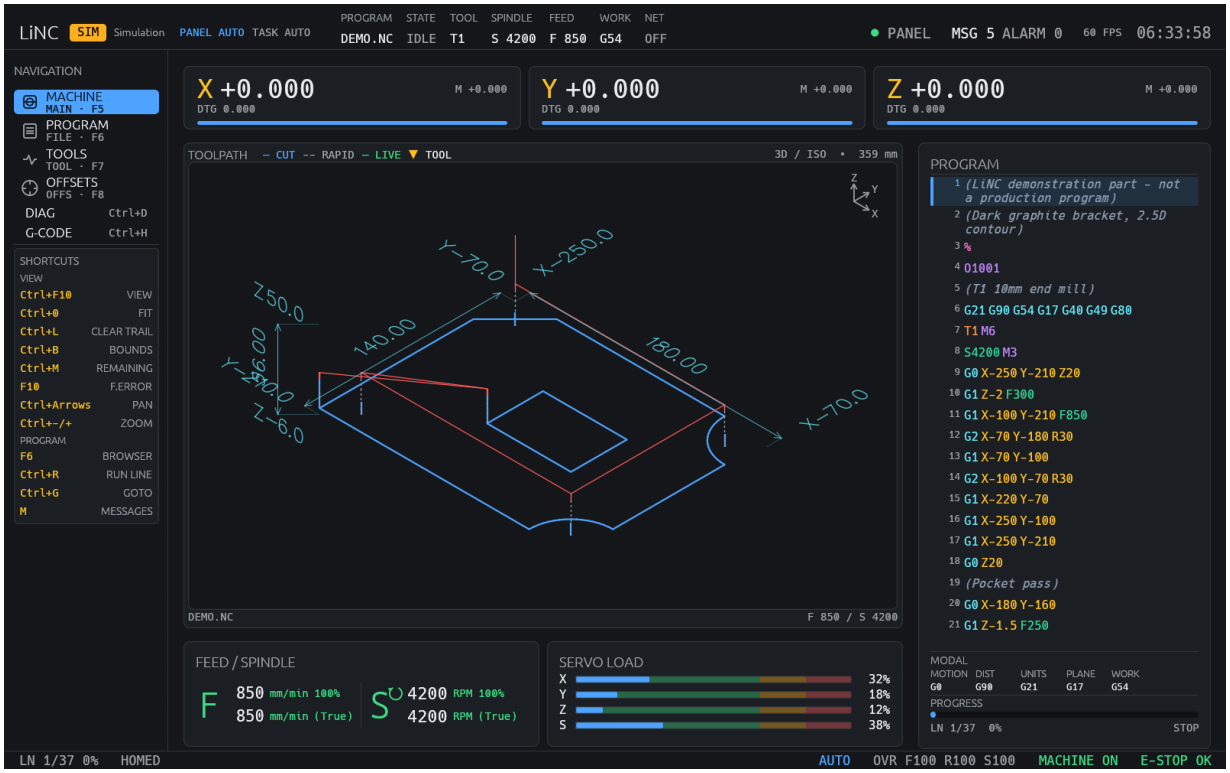


Figure 1.1: The Machine page with a program loaded: readout, toolpath, program listing, feed and spindle, servo load.

1.1 What makes LiNC different

The panel stays the panel There is no on-screen jog, no on-screen cycle start, no override slider and no software emergency stop. A second copy of a control drifts out of agreement with the hardware, so there is no second copy.

MDI is a working area The MDI pane keeps the blocks you have issued and draws the block you are typing on the toolpath before it is sent. A block that would leave the machine's travel is refused by name.

The toolpath explains itself The loaded program, the live tool trail, the machine envelope and the part bounds are separate layers with separate colours. Anything outside the machine's travel is drawn solid red.

Tool changes without MDI Select a row in the tool table and press **T**. The change and its length compensation are issued for you.

The editor checks the work The Program page is a G-code editor with syntax highlighting, undo, redo and a live preview. An edit reaches the drawing only after it parses.

Probing is guided Categories, illustrated operations, parameters and results, using the Probe Basic G-code routines that ship with LiNC.

Help is on the machine A searchable G-code reference, offline, readable while a program runs.

The network can be taken away When a program is loaded or MDI is selected, LiNC can hold the machine's external network link down, so WiFi latency cannot reach the realtime path. Ctrl + U unloads the program and releases it.

1.2 What the software will not do

These are design rules, not preferences. Each one exists so that the screen and the machine cannot disagree.

- It will not jog, home, start the spindle, change coolant or start a cycle. Those are panel actions, wired through the machine's HAL.
- It will not write an arbitrary HAL pin. It writes seven declared outputs, and the list is fixed in the profile and in the program.
- It will not save G-code, write a table, import a file or apply a probe result while the panel's PROGRAM PROTECT switch is on.
- It will not home an axis. It reports homing and explains the sequence.
- It will not guess. A stale or invalid panel state is shown as unknown and fails closed.

1.3 How to read this manual

Chapter 2 Powering up and down, emergency stop, protect switch, the network interlock.

Chapter 3 Every control on the operator panel.

Chapter 4 The keyboard, the navigation column and every shortcut.

Chapter 5 The fixed shell: top bar, bottom bar, banners, messages.

Chapter 6 Panel modes, jogging, homing and program zero.

Chapter 7 Loading, checking and running a program.

Chapter 8 MDI and the G-code editor.

Chapter 9 Tool table, tool changes, offsets.

Chapter 10 Touch probing.

Chapter 11 Diagnostics, logs and troubleshooting.

Chapter 12 Key tables and reference data.

Boxes used throughout the manual:

> **SAFETY**
>
> Instructions that prevent injury. Read them before operating the machine.

> **CAUTION**
>
> Instructions that prevent damage to the machine, the tooling or the workpiece.

> **NOTE**
>
> Context and cross references.

> **TIP**
>
> Practical hints from the shop floor.

1.4 Notation

Panel keys are drawn as CYCLE START, screen actions and labels as SAVE, keyboard keys as F6 and key chords as Ctrl + S. Lit panel lamps are drawn as ● (lit) against ○ (dark). A state word the screen paints is written as **NOT HOMED**.

Positions on the millimetre machine are in millimetres, speeds in millimetres per minute, spindle speed in revolutions per minute. A lathe profile shows X in the radius or diameter convention its INI selects.

1.5 Terms used in this manual

Term	Meaning
HMI	Human-machine interface: this program.
Panel	The operator panel, physical or simulated. It owns the machine controls.
Controller	LinuxCNC: task, motion and I/O. It is authoritative for machine state.
Task mode	LinuxCNC's own mode (MANUAL, MDI, AUTO). The panel requests it; LiNC displays it.
Panel mode	The position of the mode selector on the panel. LiNC reads it as one-hot signals and fails closed if it is invalid.
WCS	Work coordinate system, G54 to G59.3.

Term	Meaning
DTG	Distance to go: how far the tool still has to move.
MDI	Manual data input: one block, issued immediately.
HAL	LinuxCNC’s hardware abstraction layer.
Protect	The panel’s **PROGRAM PROTECT** key switch. It locks every write.
Interlock	The rule that holds the external network link down while the machine can move.

2 Power, Emergency Stop and Safety

The machine is switched on and off at the panel. LiNC follows: it shows the power and emergency-stop state on every page, and it blocks machine-modifying actions while either of them forbids motion.

2.1 The power controls

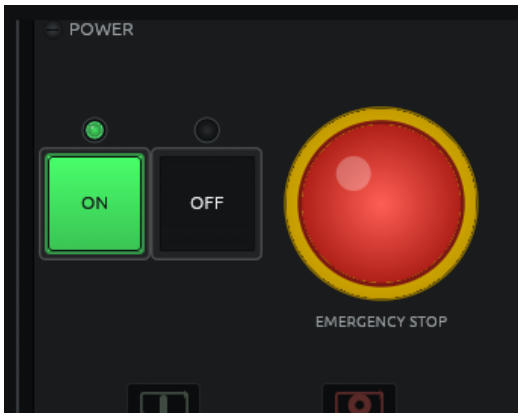


Figure 2.1: The power group: **ON** (lit green when machine power is on), **OFF**, and the emergency stop mushroom.

ON — Switches machine power on and requests *machine on* in the controller. Drives and outputs are enabled. The lamp in the key lights green.

OFF — Switches machine power off. LiNC ends its session when the controller reports that machine power has dropped.

EMERGENCY STOP — The red mushroom. It latches when pressed and is released by turning it clockwise. See section 2.3.

The simulator starts in the state a real cabinet is found in: emergency stop latched and power off. Nothing moves until the mushroom is released and **ON** is pressed. This is deliberate: the power-up sequence is part of the training.

2.2 Switching on

1. Check that the work area is clear, the guards are closed and no tool or clamp is left on the table.
2. Release the emergency stop: turn the red mushroom clockwise until it pops out.
3. Press **ON**. The controller enables the drives.
4. Wait for the panel's **READY** lamp and for the interface to show **MACHINE ON** in the bottom bar.
5. Home the axes (chapter 6). **NOT HOMED** in the bottom bar means the controller has no valid reference, and no program will start.
6. Load the program (chapter 7), check it, then run it from the panel.

> **NOTE**
>
> LiNC does not power the machine on and does not home it. Both are panel actions, wired to the controller through the machine's HAL. LiNC reports their state.

2.3 Emergency stop

> **SAFETY**
>
> The emergency stop stops all axis motion, the spindle and the coolant immediately. Keep it reachable. Never bypass it, and never work around a machine that will not leave the emergency-stop state.

Behaviour on this controller:

- **Pressing** the mushroom latches the panel input. The controller enters emergency stop, drives are disabled and motion stops. LiNC shows the red banner **EMERGENCY STOP ACTIVE — MOTION AND MACHINE ACTIONS BLOCKED** on every page, and the bottom bar reads **E-STOP ACTIVE**.
- **Releasing** the mushroom clears the latch and asks the controller to leave emergency stop. It does *not* switch machine power on: that is a separate **ON** press. This separation is deliberate, so releasing the mushroom cannot start the machine.
- After a release, press **ON** again if power dropped, check **READY**, and check the homing state before continuing.

> **CAUTION**
>
> An emergency stop interrupts the program mid-block. Before restarting, inspect the tool position and the remaining path. Use single block and a reduced feed override for the first blocks after recovery.

2.4 Switching off

1. Finish or stop the running program (**CYCLE STOP** on the panel).
2. Unload the program with **Ctrl** + **U** on the Program page if you are finished with it (section 7.8).
3. Press **OFF**. The controller shuts down and LiNC ends its session.

> **CAUTION**
>
> Do not cut the main disconnect while a program is running. Use **OFF** first, so the controller can stop in a defined state.

LiNC refuses a normal quit (**Ctrl** + **Q**) while a program is running or an axis is moving. Losing machine power is the one case where the session ends regardless, because the controller is going away anyway.

2.5 Program protect

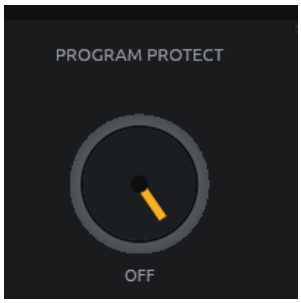


Figure 2.2: The **PROGRAM PROTECT** key switch. OFF is the working position; ON locks every write.

The panel's key switch is authoritative. While it is on, LiNC cannot:

- edit the program buffer or save it,
- write an offset, a tool row or a tool length,
- copy or import a file into the program directory,
- apply a probe result to a work offset.

The screen shows the banner **`PROTECT ON — EDITING LOCKED`** on the pages that write, and the field **`PROTECT ON`** in the bottom bar. Reading, loading and running a program remain possible: protect blocks writing, not viewing.

> **NOTE**
>
> Protect fails closed. If the switch state is unknown or the panel data is stale, LiNC treats protect as *on*. Every write checks it again in the backend; a value the screen sampled earlier is not enough.

Unsaved edits are not thrown away when protect is switched on. The buffer stays, keeps its dirty marker and becomes read-only until the switch returns to OFF.

2.6 Limits override

If an axis stands on a limit switch, the panel's jog keys are refused. The keyboard chord **Ctrl** + **Shift** + **O** toggles the controller's limit-switch override so the axis can be jogged off the switch.

> **CAUTION**
>
> While the override is on, the limit switches are ignored and the machine can be driven past them. The screen shows a blinking **`LIMITS OVERRIDE ON`** in the bottom bar and the banner **`LIMITS OVERRIDE ACTIVE — LIMIT SWITCHES ARE IGNORED`**, but neither stops the machine. Jog off the switch, then switch the override off again.

2.7 Panel offline and invalid mode

LiNC reads the panel through the panel service. Two failures are shown as banners and both fail closed:

`PANEL OFFLINE`	No current panel data. Machine-modifying actions are blocked, and the top bar's **`PANEL`** dot turns red.
`PANEL MODE INVALID`	The mode selector reports zero or more than one position. LiNC does not guess: the mode reads **`UNKNOWN`** and mode-dependent actions are refused until a valid position arrives.

The hardware emergency stop is independent of both. It works even when the panel service is down, because it is wired in the machine's HAL.

2.8 The external link interlock

On a machine whose only external network link is WiFi, LiNC can hold that link down while the machine is able to move. WiFi association, driver interrupts and packet bursts all cost realtime, and none of them are needed while a program runs.

> **NOTE**
>
> The interlock only ever touches the external link named in the profile. The machine's internal fieldbus is never touched.

When it is armed. The link is held down while a program is loaded or the panel is in `MDI`. It is released when nothing is loaded and no MDI block is being set up. This is arm-on-intent, not arm-on-motion: a loaded program that is paused still holds the link down, because it can be resumed.

The `NET` field. The top bar always shows the link state:

Value	Colour	Meaning
DOWN	green	Link down. This is the state the interlock wants while armed.
UP	muted	Link up, no program motion. Normal while setting up.
UP	amber	Link up *and* program motion. The one state to look at.
MEAS	accent	A link measurement is running.
?	amber	State unknown or stale. Fails closed.
OFF	muted	No interlock configured in this profile.

Amber is reserved for program motion on purpose. Jogging and homing keep the link up by design, and colouring every setup jog amber would train the operator to ignore the one colour that matters.

Releasing the link. Press **Ctrl** + **U** on the Program page. LiNC closes the program at the controller, resets the interpreter, re-reads the program name and only then reports success. The **NET** field returns to **UP** within a few seconds, after the radio has associated and a default route exists. A toast says **UNLOAD REQUESTED — WATCH NET**.

> **CAUTION**
>
> Leaving a program loaded keeps the link down, and the link is the only remote way into the machine. If you need remote access while nothing is being cut, unload the program first. Profiles that must keep the link up can set `keep_up = true` in `[network]`.

If the link is up while a program is loaded, commands that would run or change the program are refused with **NETWORK BUSY** rather than being sent.

2.9 Probe trip

If the probe input rises during a move that is not a probe, the controller aborts the move and reports an operator error. LiNC treats this as a latched fault, not a passing message:

- A red banner carries the controller's own text on every page, for example **PROBE TRIPPED — ... — MACHINE SWITCHED OFF. MOTION AND MDI BLOCKED. RESET (ESC) TO CLEAR.**

- The machine is switched off through `linc.machine-off`, so the panel cannot jog into the probe either. One exception exists: a trip raised with the program paused and nothing moving leaves the machine on, because nothing had to be stopped.
- MDI, probe, tool-change, jog-parameter and table writes fail closed until the trip is cleared.
- **Esc** (or the banner's **RESET**) clears it, from any page. That sends the controller reset which clears the motion error. Re-powering afterwards needs a fresh **ON**.

2.10 Safety rules, in short

The safety and liability disclaimer that applies to this software is printed at the front of this manual, before the contents.

SAFETY

- The panel is the only place machine motion starts. Do not look for a jog or a cycle start on the screen.
- Release the emergency stop, then press **ON**: two separate actions.
- Never leave the machine in limits override.
- Never run a program that shows red path outside the machine envelope without knowing why.
- Unload the program before relying on the network link.

3 The Operator Panel

The operator panel is the machine's control surface. Every action that moves the machine, changes the spindle or changes an override starts here. LiNC reads the panel and shows what it is doing; it never sends those commands.

The panel runs as its own process (`linc-panel`) and owns the panel link. It publishes a fixed set of button, lamp, display and override signals to the controller's HAL, named `cnc_panel.*`. The interface reads the same signals. A crash of the interface therefore does not take the panel away, and the interface never opens the panel port.

> **NOTE**
>
> The interface cannot write a `cnc_panel.*` pin. It writes only its own seven declared outputs, and `net-safe` is the only one that reaches machine logic.



Figure 3.1: The simulated operator panel, mill face. A physical panel carries the same groups; the machine builder decides the exact legends of the machine function keys.

3.1 Power and emergency stop

The power group and the emergency stop are described in chapter 2. The panel is where they are pressed; the screen only reports them.

3.2 The mode selector



Figure 3.2: The **MODE** selector. Its eight steady positions are read as one-hot signals: exactly one position must be active.

The selector is read as a steady position, not as pulses. Zero or several active positions is a fault: LiNC shows **`PANEL MODE INVALID`** and refuses mode-dependent actions.

Position	Panel / controller	What LiNC does
EDIT	Manual mode, editing allowed	Opens Program in the **CODE** layout and focuses the editor. Saving is allowed only here.
AUTO	Automatic mode	Brings the Machine page forward. Cycle start then belongs to the panel.
MDI	Manual data input	Opens the Machine page with the MDI pane in place of the program card. **Enter** issues the block.
JOG	Manual mode, continuous jog	Opens the Manual page: live toolpath, coordinates, jog context.
INC	Manual mode, step jog	Same page, with the increment row. The steps come from the machine INI.
MPG	Manual mode, handwheel	Same page; the handwheel selects its own step.
HOME	Manual mode, homing	Opens the Homing page with the machine position and the homing state of each axis.

Position	Panel / controller	What LiNC does
PROG.HOME	Manual mode, program zero	Opens the Homing page titled **PROGRAM ZERO**. The axis keys send the program-zero blocks, not jogs.

> **NOTE**
>
> The selector is authoritative for what the panel keys do. LiNC's own page changes are a convenience copy of that state; where the two disagree, the panel wins. The top bar shows **PANEL** and **TASK** separately, so a disagreement is visible.

3.3 The override knobs

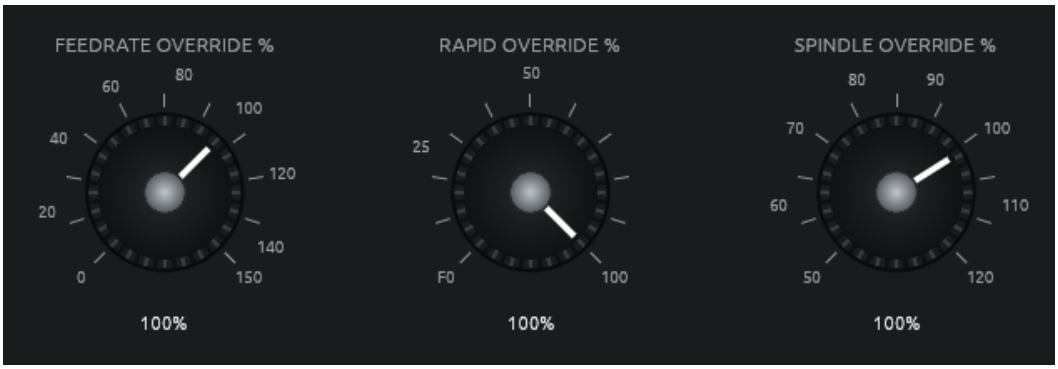


Figure 3.3: Feedrate, rapid and spindle override. The percentage under each knob is the current value; the interface repeats all three in the bottom bar.

The three knobs are steady analogue values sampled by the panel service. LiNC displays them as **OVR F100 R100 S100** in the bottom bar, and the Machine page uses the feed and spindle percentages in its **FEED / SPINDLE** card.

- **Feedrate override** scales programmed feed. On the shipped panel it ranges from 0 to 150 %.
- **Rapid override** scales rapid moves. `F0` stops rapids.
- **Spindle override** scales the commanded spindle speed. It does not change the displayed true speed, which comes from feedback.

> **CAUTION**
>
> An override of 0 % feed looks like a machine that will not move. Check the knobs before diagnosing a fault: the bottom bar always shows them.

3.4 Cycle keys and axis keys



Figure 3.4: Left: **CYCLE START** (illuminated white) and **CYCLE STOP**. Centre: the axis lamps. Right: the alarm lamps.

CYCLE START — Starts the loaded program. The controller gates it: machine on, no emergency stop, axes homed, interpreter idle, and the panel in AUTO. LiNC does not issue it and cannot issue it.

CYCLE STOP — Stops the program and stops motion.

Axis keys — Jog in JOG and INC, home in HOME, and send the program-zero block in PROG.HOME, all according to the mode selector and the machine's HAL wiring.

RAPID — With an axis key, jogs at rapid rate; in HOME it starts homing all axes; in PROG.HOME it sends the all-axes program-zero block.

The axis lamps light for the axes the controller reports. The alarm lamps report machine conditions (machine ready, run, and the machine's own alarm set). **READY** follows machine power and an idle or paused program.

3.5 The displays

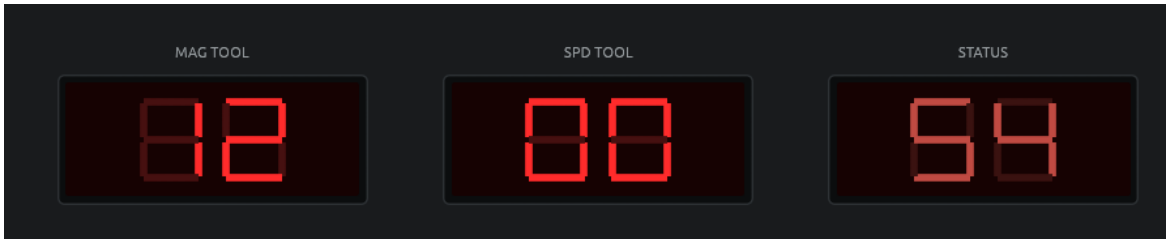


Figure 3.5: The tool displays and the two-digit status display. Here the status display shows 54, the active work coordinate system.

- **MAG TOOL** is the tool in the spindle as the controller reports it.
- **SPD TOOL** is the prepared tool, where the machine has a changer.

- **STATUS** shows the active work coordinate system, `54` to `59`. It shows `--` when the value is unknown, stale or outside those systems, and never a guess.

The interface writes `linc.status-display` and the machine's post-GUI HAL maps it to the display. This is one of the seven declared outputs.

3.6 The machine function keys

The key matrix carries the machine functions: spindle forward, stop, reverse and orient; coolant; air blow; chuck; lube; chip conveyor; work lamp; safe door; tool changer and pot keys; and the four free keys **K1** to **K4**. Each key is a HAL button; the machine configuration decides what it switches.

> **NOTE**
>
> Only the keys the machine wires do something. A key with no net in the machine's HAL is inert, and the simulated panel greys the keys its profile does not wire. Nothing on the screen duplicates these functions.

3.7 The simulated panel

The simulated panel is a window that speaks the same protocol as a physical panel. It is used for the simulator, for training and for a machine that has no panel cabinet.

- It draws the face the profile selects: mill or lathe, with that face's keys, lamps and displays.
- It publishes the same `cnc_panel.*` signals as the real driver, so the interface and the HAL cannot tell the difference.
- It starts in the safe state: emergency stop latched, power off.
- It runs on a second monitor if you put it there. With no physical panel, a touch screen or a mouse can operate it; that makes the pointer a replacement for the panel hardware, not a second set of machine controls.

The panel window also answers the keyboard, so it can be used without a pointer:

Key	Action
P	Power on (also releases the emergency stop)
O	Power off
E	Press / release the emergency stop
R	Rapid (hold)
C	Cycle start
S	Cycle stop
1 … 8	Mode selector: EDIT, AUTO, MDI, JOG, INC, MPG, HOME, PROG.HOME

TIP

In a demonstration on one computer, keep the panel window on the second monitor and give the interface the first one. The interface's F5 to F8 keys and the panel's number keys act on different windows, so the focus must be where the key belongs.

4 The Keyboard and Navigation

LiNC is built for a CNC panel keyboard: the keys are labelled with what they do at the machine, not with the letters a PC keyboard happens to send. A normal PC keyboard works too, and the profile can name the keys either way: `[ui] keyboard = "panel"` prints the panel legends, `[ui] keyboard = "pc"` adds the keys a PC keyboard sends.

Every action in the interface can be reached from the keyboard. Nothing needs a pointer, nothing needs a touch, and nothing depends on a **Tab** key.

4.1 The navigation column

The main layer

The main layer holds what a program is typed from:

- the address letters `O N G X Y Z F D H M S T P Q R` and all ten digits;
- ENT to commit, BkSp to delete, `Space`;
- the punctuation a program needs: `. - / ;` and their shifted forms;
- the page and cursor keys, and `Shift` and `Ctrl`;
- the five machine keys **MAIN**, **FILE**, **OFFS**, **TOOL**, **WIN** and **ESC**.

The letters that are rare in G-code – `A B C E I J K L U V W` – are not on the main layer. They are the blue legends, reached with **Shift** or a double tap.



Figure 4.1: The navigation column and the shortcut pane. The shortcut pane lists the keys that work on the page in front; it is a reminder, not a set of buttons.

Row	Key	Opens
MACHINE	F5	The Machine page, or in EDIT/JOG/INC/MPG/HOME the page that belongs to the panel mode.
PROGRAM	F6	The Program page. Pressing it again opens the Files browser; pressing it in Files brings the program back.
TOOLS	F7	The tool table.
OFFSETS	F8	The work offsets. Pressed again, it moves on to the probe tabs, one press per tab, and from the last tab back to Offsets.
DIAG	Ctrl + D	Diagnostics.
G-CODE	Ctrl + H	The G-code help.

The profile may hide rows or change their order, so a machine can ship without the pages it does not use.

4.2 Focus and movement

The shell has three focus regions: the navigation column, the page, and the page's actions. The focused element always has a visible ring.

Arrow keys Move within the current region: rows, table cells, list entries, chips.

Enter Activates: open a folder, load a program, commit a cell, issue an MDI block, run the highlighted routine.

Esc Goes back: closes a dialog, leaves the help article, cancels an edit. On any page while a program is active it resets the program.

Home / **End** First and last entry of a list. Inside a text field they move to the start and end of the line.

PgUp / **PgDn** A screenful in a list or a read-only program; on Diagnostics and the probe tabs, the next tab.

A text field that holds the keyboard keeps the page keys for itself: inside the editor, **PgUp** moves the cursor rather than the page.

4.3 Keys that work everywhere

Keys	Action
F5 … **F8**	Navigation, as in the table above.
Ctrl + **D**	Diagnostics.
Ctrl + **H**	G-code help.
M	The full-screen messages dialog.
Esc	Back, cancel, or reset the program when one is active.
Ctrl + **Q**	Quit. Refused while a program runs or an axis moves.
Ctrl + **O**	Open the Files browser, from any page except Program, where **F6** does it.
Ctrl + **S**	Save the editor buffer.
Ctrl + **Z** / **Ctrl** + **Y**	Undo / redo in the editor.
Ctrl + **F**	Search the G-code help (Help page only).
Ctrl + **Shift** + **O**	Toggle the limit-switch override.

> **CAUTION**
>
> Ctrl + Q ends the session. It is refused while the machine is busy on purpose: a quit prompt in the middle of a cut is a dialog in the way.

4.4 Keys for the toolpath

Every page that draws a toolpath answers the same view keys, so a view command learned on one page works on the next.

Keys	Action
Ctrl + F10	Cycle the view: ISO, X, Y, Z. A lathe toggles ISO and its XZ view.
Ctrl + 0	Fit the drawing to the page.
Ctrl + L	Clear the live tool trail only.
Ctrl + B	Cycle the bounds: program dimensions, machine envelope, program box, off.
Ctrl + M	REMAINING: show only what is still to run.
Ctrl +arrows	Pan the view.
Ctrl + + / Ctrl + -	Zoom in / out.

The pointer does the same: dragging pans, the wheel zooms, a middle-drag orbits in 3D, and a double click returns to the default isometric view. On the keyboard, Ctrl + F10 and Ctrl + 0 cover the two commands that matter most.

4.5 Keys for programs and the editor

Keys	Action
Ctrl + R	RUN FROM LINE: start the loaded program at the selected line.
Ctrl + G	GO TO LINE: ask for a line number and move the view there.
Ctrl + U	UNLOAD: close the program and release the network interlock.
F10	Toggle the Program layout: SPLIT and CODE.
Ctrl + ↑ / Ctrl + ↓	Move the selected line up or down in the editor.

Keys	Action
Ctrl + ← / Ctrl + →	Move by word in the editor and the MDI input.
Ctrl + A / C / X / V	Select all, copy, cut, paste in the editor.
Ctrl + K	Delete the current line in the editor.
Ctrl + D	Duplicate the current line in the editor. It shadows the Diagnostics key while the editor has the keyboard.

> **NOTE**
>
> Ctrl + R has three page-local meanings: run from line on Program, run the selected block in the MDI pane, and refresh the listing in Files. The shortcut pane always shows the one that applies now.

4.6 Reading the shortcut pane

The pane under the navigation column lists the page's own keys in groups: **VIEW**, **PROGRAM**, **BROWSE**, **FILE**, **MEDIA**, **LOG VIEW**, **PROBE KEYS** and so on. Each line is a key or chord on the left and the action it performs on the right. The list changes with the page and with the state: while the MDI pane is up it lists the MDI keys instead of the program keys, and while the editor holds the keyboard it lists the line-move chords.

When the pane shows nothing, the page has no page-local keys. The global keys above still work.

4.7 Dialogs

A dialog owns the keyboard while it is open. Enter confirms, Esc cancels, the arrows move between its buttons or fields, and nothing behind it reacts to a key. The messages dialog scrolls with PgUp / PgDn and closes with Enter or Esc.

The unsaved-changes dialog has no third option: save, discard, or cancel the action that would have thrown the buffer away. Saving is greyed out when a condition blocks it, and the dialog names the condition.

4.8 The CNC keyboard

The machine keyboard is a 48-key USB keyboard with per-key RGB backlight. Its layout is the industrial-controller layout: the letters and digits a G-code program is written from sit on the main layer, and everything else is reached with **Shift** or by tapping a key twice. The bottom row carries the navigation keys, so the whole interface is within one hand's reach.

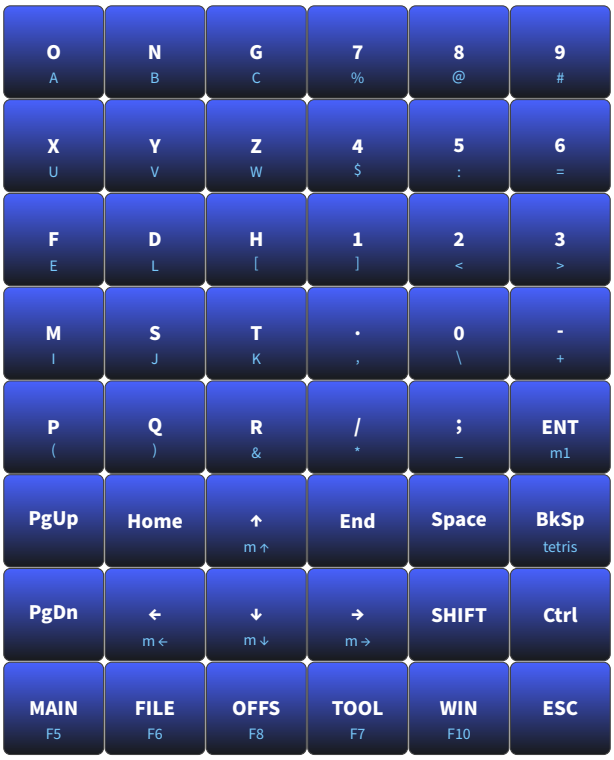


Figure 4.2: Keyboard layout. Centre: the main layer. Small blue legends: the shift and double-tap layer (`m` = mouse key).

Shift and double tap

Shift — Hold it, then press the key: the blue legend is sent. The dual-layer keys glow brighter blue while Shift is held.

Double tap — Tap the same key twice within 0.3 s. The first character is erased and the blue legend is written instead: **O** **O** writes A, **7** **7** writes %, **F** **F** writes E.

Letters are transmitted lowercase. G-code words are not case sensitive in LinuxCNC, so lowercase letters are fine in MDI and in the editor.

A handful of secondary legends are not characters: the arrow keys become a mouse (`m↑` moves the pointer, `m1` on **ENT** clicks), and a double tap on **BkSp** starts a small game (section 4.8).

Backlight

An idle key glows dim blue. A pressed key lights **white**; when the press produces a blue legend, it lights **yellow**. The backlight is the state display of the keyboard: if a key stays yellow after it was pressed, the secondary layer was sent.

The machine keys

Legend	Sends	Opens
MAIN	F5	The Machine page, or in EDIT, JOG, INC, MPG or HOME the page that belongs to the panel mode.
FILE	F6	Program. Pressed again, Files; in Files, back to Program.
TOOL	F7	Tools.
OFFS	F8	Offsets. Pressed again, the probe tabs.
WIN	F10	SPLIT / CODE on Program. With Ctrl, cycle the toolpath view.
ESC	Esc	Back, cancel, or reset the program when one is active.

The interface does not depend on the printed letters. It depends on the keys these legends send, which is why a normal PC keyboard works as well: F5 to F8, F10 and Esc are the same keys.

Typing in the editor

The keyboard has no Tab and no Delete. Write spaces, delete with BkSp, and use Ctrl + A, Ctrl + C, Ctrl + X and Ctrl + V to select, copy, cut and paste. Line-based editing – duplicate a line, delete a line, move a line – is on Ctrl + D, Ctrl + K and the Ctrl + arrows chords, because those do not need a pointer.

Service combinations

> ⚠ **CAUTION**
>
> Shift + MAIN puts the keyboard into its firmware bootloader, and Shift + ESC quits the user interface. Both are service functions. Do not press them during machining.

TIP

Hold **Shift** + **BkSp** for three seconds and the LED matrix plays a small game (arrows steer, hold again to quit). A progress bar on the bottom row shows the activation. Best used during coffee, not during cutting.

NOTE

If a key seems dead, look at the shortcut pane first: the key may be page-local and not belong to the page in front. **Ctrl** + **D** inside the editor is a good example – it duplicates a line there, and reaches Diagnostics everywhere else.

5 The Screen

The screen has a fixed shell. Every page keeps the same frame, so the numbers an operator watches are always in the same place.



Figure 5.1: The top bar. Left: program identity and the panel and task modes. Centre: the machine metrics. Right: panel link, message and alarm counters, frame rate and clock.

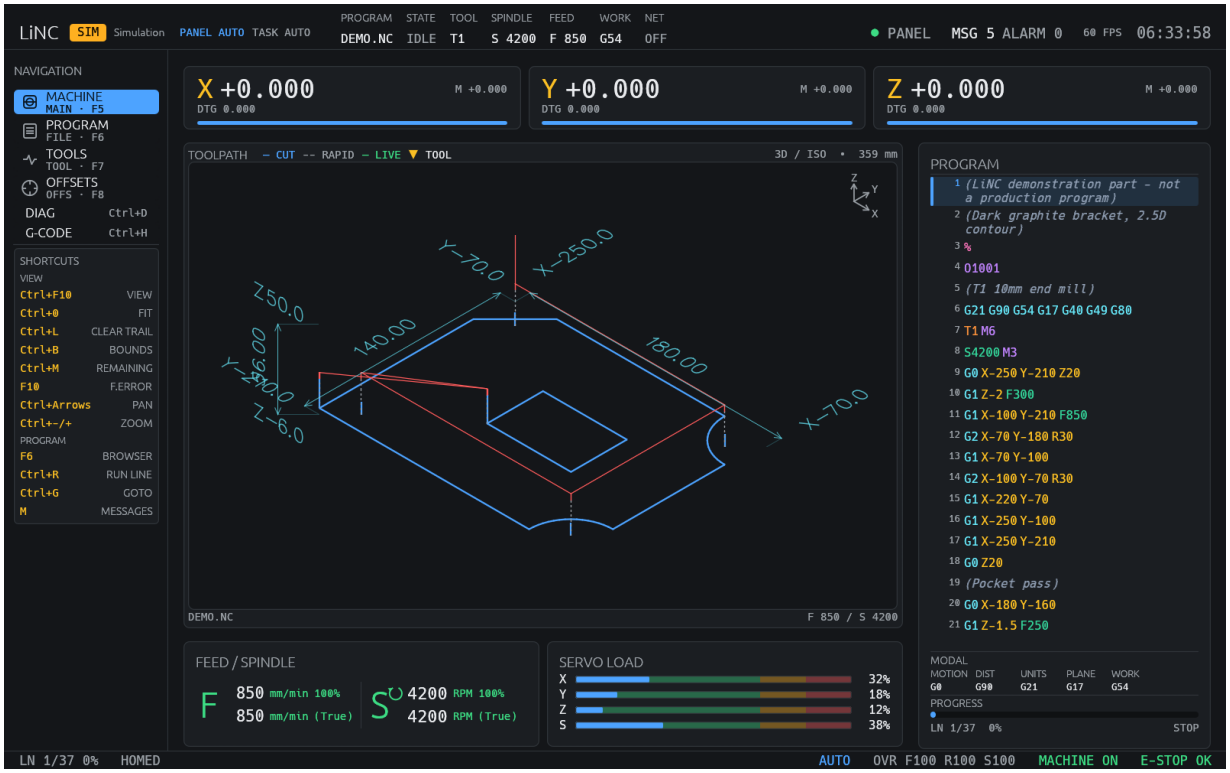


Figure 5.2: The Machine page. The shell around it – top bar, navigation column, shortcut pane, bottom bar – is present on every page.

5.1 The top bar

LiNC The program name, and a **SIM** badge when the session is a simulation. The machine name comes from the profile.

PANEL	The panel's mode selector position. Accent-coloured when valid, red when the mode is unknown or invalid.
TASK	The controller's own task mode. It follows the panel, but it is the controller that decides.
PROGRAM	The loaded program's file name, or `--` when none is loaded.
STATE	**`IDLE`**, **`RUNNING`** or **`PAUSED`**, with the colour that belongs to it.
TOOL	The tool in the spindle as the controller reports it.
SPINDLE	The measured spindle speed in revolutions per minute.
FEED	The programmed feed rate.
WORK	The active work coordinate system.
NET	The external link state (chapter 2).

On the right: the **`PANEL`** status dot with its link state, the message and alarm counters, the render rate and the clock. A frame rate below 30 is amber: it is the first number to ask about when the interface feels slow.

> **NOTE**
>
> The counters are read-only. **`MSG`** counts the entries in the message log, **`ALARM`** counts the errors among them. Neither is a button.

5.2 The bottom bar



Figure 5.3: The bottom bar: machine state on the left in the flow, the read-only safety and override group on the right.

Fields appear only when they have something to say:

Field	Meaning
`LN 12/340 34%`	The executing line, the line count and the program progress.
`HOMED`	Every axis is homed. A partly homed machine says **`HOMED X`** and names the axes it already trusts.
`NOT HOMED`	No axis is homed. Motion commands are refused.
`LIMIT X`	A limit switch is held. Motion is blocked and the axis is named.
`TC T7 PENDING`	A manual tool change is waiting for confirmation.

Field	Meaning
DTG Z +1.234	Distance still to go on the axis that is moving. On the Program page this becomes a full coordinate line for every axis.
E-STOP ACTIVE / **E-STOP OK** / **E-STOP ?**	The emergency-stop state, or a question mark when the data is stale.
MACHINE ON / **MACHINE OFF**	Machine power. **MACHINE OFF** blinks: it is a fault, not a normal idle.
PROBE READY / **PROBE TRIPPED** / **PROBE ?**	The wired probe input. The field is drawn only while the probe tool is in the spindle.
PROTECT ON	The panel's program-protect switch is on.
OVR F100 R100 S100	Feed, rapid and spindle override.
LIMITS OVERRIDE ON	Blinking. The limit switches are being ignored.
AUTO	The panel mode, repeated here for a glance.

The bar carries no buttons. It is the one line that stays visible while a program runs, so every action on it is an action the operator did not intend.

5.3 Banners

Banners sit under the top bar and cannot be dismissed by changing page. Each one is a state that must not be hidden:

- **EMERGENCY STOP ACTIVE — MOTION AND MACHINE ACTIONS BLOCKED**
- **LIMITS OVERRIDE ACTIVE — LIMIT SWITCHES ARE IGNORED**
- **PANEL OFFLINE — MACHINE ACTIONS BLOCKED**
- **PANEL MODE INVALID — FAILS CLOSED**
- **PROBE TRIPPED — ...** with the **RESET (ESC)** action
- **PROTECT ON — EDITING LOCKED** on the pages that write
- an external-change warning on Program when the file on disk changed under the buffer

5.4 Messages

The controller is the only thing that can say why a program stopped. Its operator errors arrive on the controller's error queue and become alarms: a row in the message list, an entry in the messages dialog, a count in the top bar and a red toast.

- A one-line strip above the bottom bar shows the newest entries, newest first. It disappears when the controller has said nothing.
- **M** opens every message, full screen, with the time, severity, source, code and text. **PgUp** / **PgDn** scroll, **Enter** and **Esc** close.
- A status note lives for five seconds; an error lives for twelve, because the operator is still reading the console when it appears.
- Identical texts inside a one-second burst are one entry. A repeat after that window is the operator trying again and is reported again.

> **TIP**
>
> When something behaves unexpectedly, press **M** before touching anything. The last controller error usually names the block, the axis or the missing homing reference.

5.5 The pages

Page	Opened by	Contents
Machine	**F5**, or AUTO	Readout, toolpath, program card, feed and spindle, servo load; the MDI pane in MDI mode.
Program	**F6**, or EDIT	G-code editor and toolpath in SPLIT or CODE.
Files	**F6** again, or **Ctrl** + **O**	USB media and the program directory.
Tools	**F7**	Tool table, tool changes, tool length, optional Loobric sync.
Offsets	**F8**	Work offsets and their preview.
Touch probe	**F8** again	Probing categories, parameters, results.
Manual	JOG, INC, MPG	Live toolpath and coordinates; the panel does the jogging.
Homing	HOME, PROG.HOME	Homing state and sequence; the panel does the homing.
Diagnostics	**Ctrl** + **D**	Health, logs, HAL and NML detail.
G-code help	**Ctrl** + **H**	Offline G-code reference.

The manual page and the homing page are *mode-driven*: the panel selector opens them and leaving the selector position closes them again. They are not navigation rows.

5.6 The G-code help

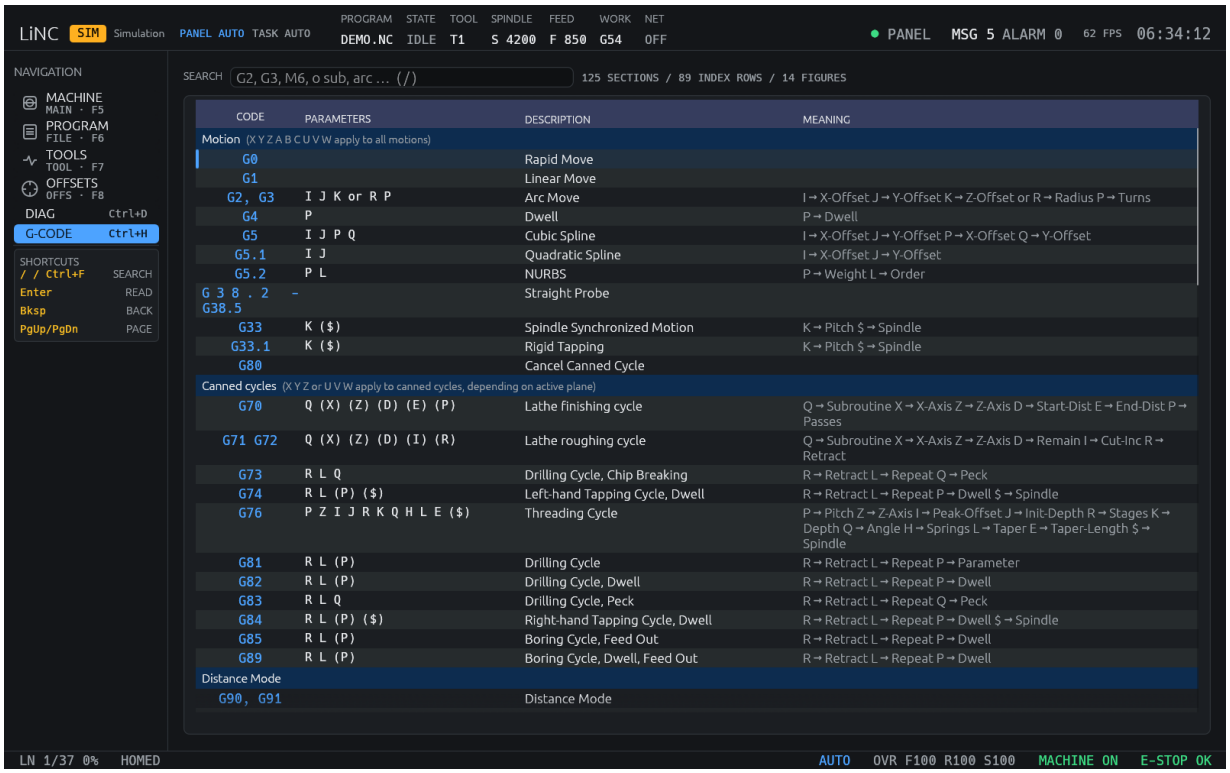


Figure 5.4: The G-code help: the quick-reference table with a code selected, and its parameters in the columns the reference uses.

Ctrl + **H** opens an offline LinuxCNC G-code reference: one table grouped by motion, canned cycles, M-codes, O-words and other codes, with the parameters, a description and the meaning of each code.

- **/** or **Ctrl** + **F** focuses the search. Matching ignores spaces and commas, so `g2g3` finds `G2` , `G3`.
- **Enter** opens the full entry in a near-fullscreen dialog: description, parameter table and notes. Arrows and page keys scroll it.
- **Backspace** returns through the links you followed; **Esc** closes the entry, or clears the search when no entry is open.
- The dialog has its own warning, note, tip and caution blocks, so a caution does not read like a footnote.

The reference ships with the program. It works with no network, and it stays readable while a program runs.

5.7 Things the screen will not show

There is no on-screen jog, no cycle start, no feed-hold, no override slider, no spindle key and no software emergency stop. If a page appears to offer one of them, it is a readout:

- The Machine page's **`RUNNING`** / **`PAUSED`** / **`STOP`** word is the program card's status line, not a stop button.
- The progress bar has no click target.
- The DRO cards print values; the editing fields that do accept numbers are on the Tool and Offset pages and the probe parameter card.

6 Modes, Jogging and Homing

The panel's mode selector decides what the axis keys do and which page the interface puts in front. The interface never changes the panel's mode and never jogs: it follows.

> **NOTE**
>
> Every position on the selector is described in section 3. This chapter describes what the operator does in each of them.

6.1 EDIT, AUTO and MDI

EDIT Selects the Program page in the **CODE** layout with the editor focused and editable. Saving is only possible here.

AUTO Brings the Machine page forward. Cycle start is the panel's key; the interface shows the line, the progress and the toolpath.

MDI Opens the Machine page with the MDI pane in place of the program card. See chapter 8.

Leaving EDIT with an unsaved buffer raises a warning once, because the controller refuses a save outside EDIT and AUTO can therefore only run the loaded file. The dialog offers **DISCARD BUFFER** and **KEEP**.

6.2 JOG: continuous jogging

- Hold an axis key on the panel. The axis moves while the key is held.
- **RAPID** with an axis key moves at rapid rate; the rapid lamp lights.
- The jog speed is the profile's `[jog] max_speed` (or `rapid_speed` with RAPID), scaled by the feed override. It is a machine setting, not a screen setting.
- The Manual page shows the live position, the trail and the loaded program path while you jog. It offers no jog buttons.

6.3 INC: step jogging

Each key press moves exactly one increment. The increments are the machine INI's `[DISPLAY] INCREMENTS` list, so the interface offers exactly the steps the controller was configured with.

- The increment row appears on the Manual page in INC mode.
- **1** to **9** select the step that carries that number. The list's leading `JOG` entry is continuous jog, which is a mode, and is not offered as a step.
- The selection goes to the controller as `linc.jog-increment` and is remembered across sessions.

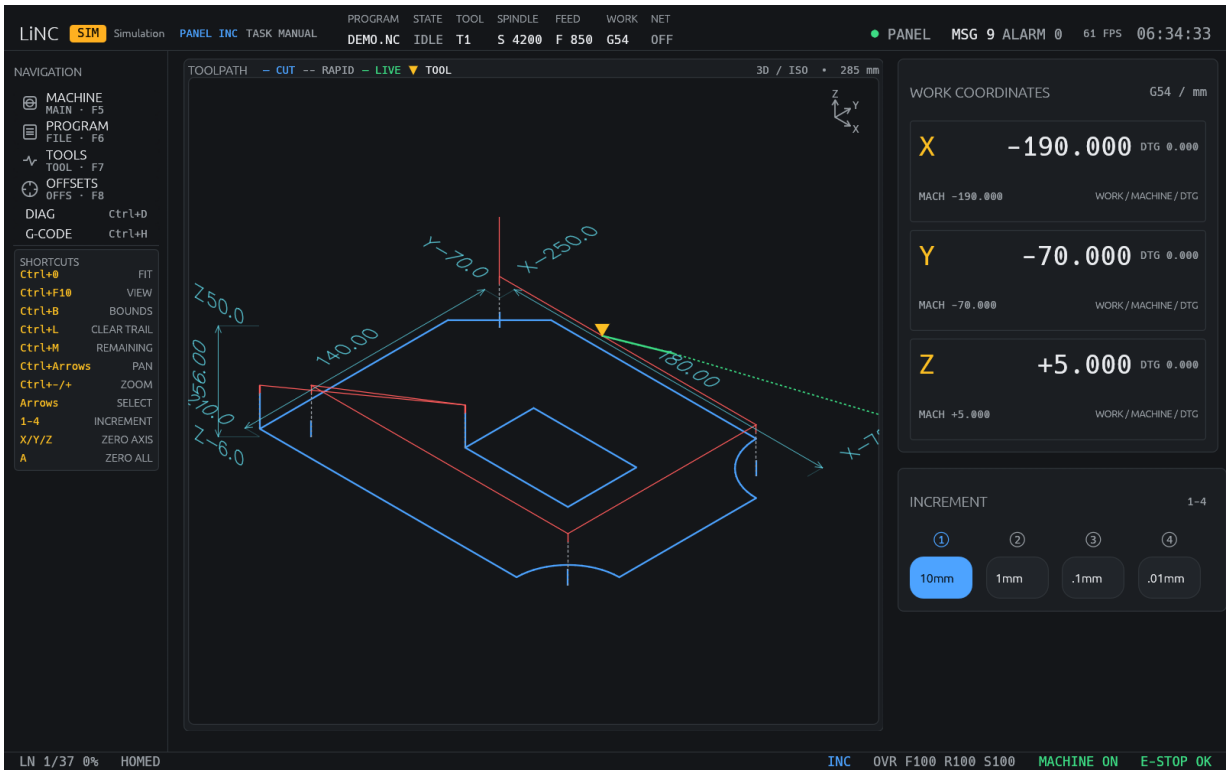


Figure 6.1: The Manual page in INC mode. The increment row appears under the coordinates, because the step moves those coordinates.

6.4 MPG: the handwheel

MPG enables jogging from the machine's handwheel. Rotate the dial to move the selected axis; the handwheel's own keys select the axis and the step. The Manual page shows the mode, the axis selection, the coordinates and the trail.

> **NOTE**
>
> The handwheel is a jog input only. Its mode, cycle, spindle, coolant, override, program and macro keys are not wired into the interface or the controller's control paths. Outside MPG its jog input is ignored as well.

6.5 Zeroing an axis while setting up

In JOG, INC, MPG, HOME and PROG.HOME the Manual and Homing pages accept touch-off keys. They write the work offset; they do not move the machine.

Key	Action
X / Y / Z	Set the active work coordinate of that axis to zero
A	Set every axis to zero
O	Toggle the limit-switch override

The keys are refused when the machine is off, an axis is not homed, the interpreter is busy or program protect is on. A means something different on other pages; see chapter 12.

6.6 HOME: homing

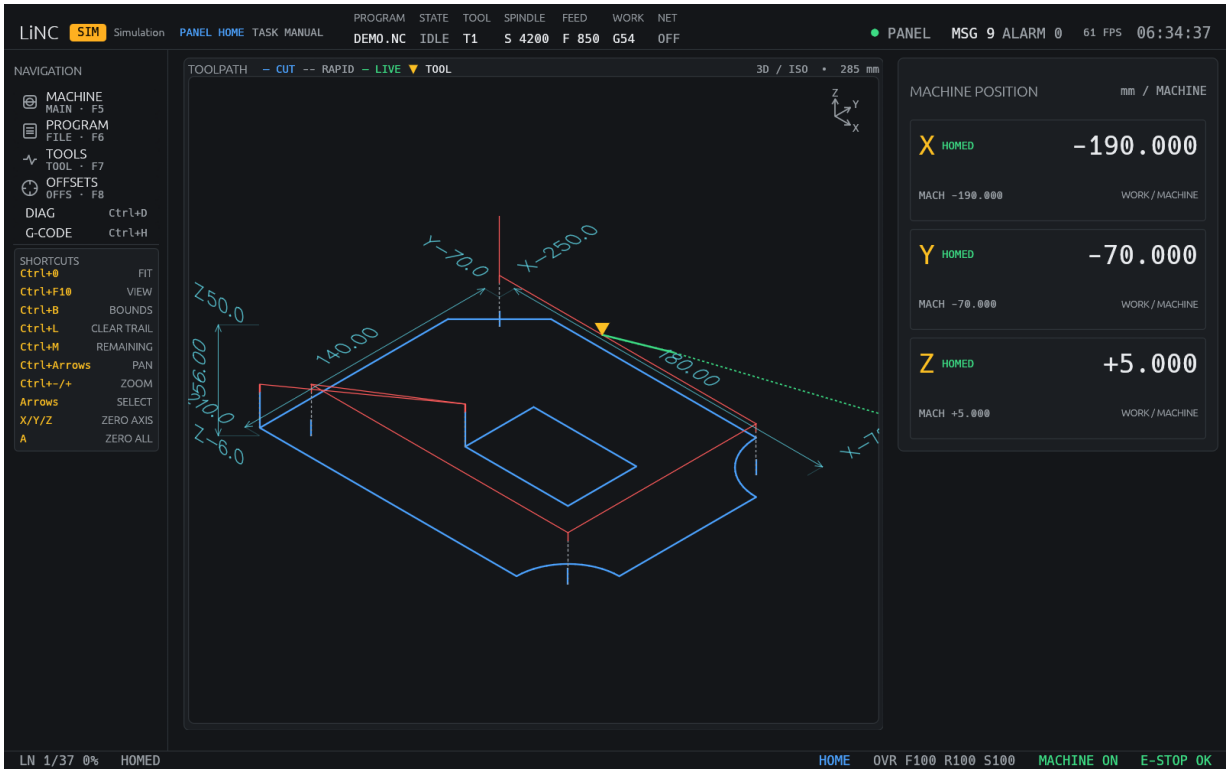


Figure 6.2: The Homing page in HOME mode: machine position with a homing badge per axis, and the same toolpath as the Manual page.

Homing establishes the machine reference. It is required after power-on and after a loss of reference. Which key homes which axis is the machine's HAL wiring, and the page documents it.

1. Turn the selector to **HOME**. The Homing page opens.
2. Press **RAPID** to home all axes, or the axis key of a single axis.
3. Each axis row shows a badge: **HOMED**, **HOMING**, **NOT HOMED**, **UNKNOWN** or **NOT CONFIGURED**.
4. The bottom bar changes from **NOT HOMED** to **HOMED** when every axis is home.

The homing order and the search, latch and index settings come from the machine INI and are shown on the page. With `NO_FORCE_HOMING = 0`, no program and no MDI block runs before the axes are homed.

> **CAUTION**
>
> Keep a hand near **CYCLE STOP** while homing. Check that the table is clear of clamps and tools before Z moves.

> **NOTE**
>
> LiNC does not home an axis. The panel's home keys drive the controller's homing input directly, which is why homing keeps working when the interface is busy or restarting.

6.7 PROG.HOME: returning to program zero

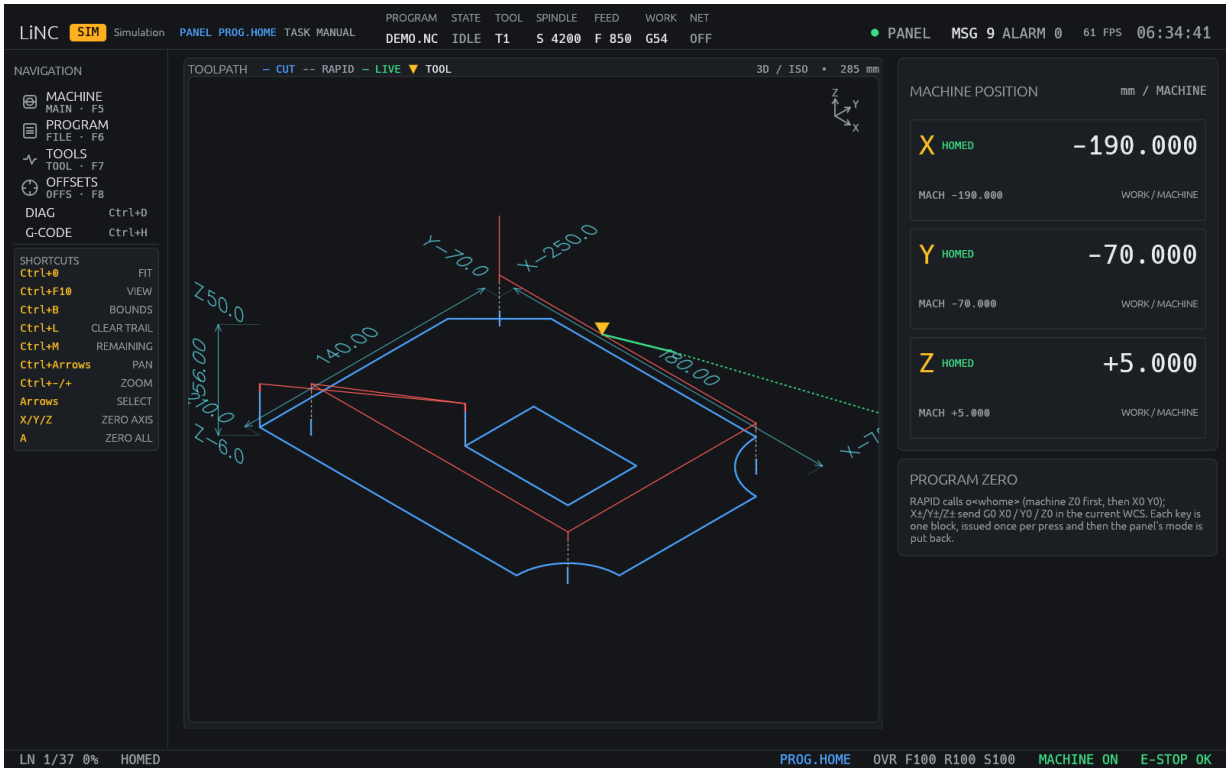


Figure 6.3: The same page in PROG.HOME mode. The readout is headed **PROGRAM ZERO**; the panel keys send program-zero blocks.

In PROG.HOME the axis keys are not jogs. Each press sends one block, and the interface issues it:

Key	Block
RAPID	`o<whome> call`: machine `Z0` first, then work `X0 Y0`
X+ / X-	`G0 X0`
Y+ / Y-	`G0 Y0`
Z+ / Z-	`G0 Z0`

- One block per press: a held key does not repeat.

- The blocks move to the program zero of the *active* work coordinate system. This is not machine home and not G53.
- The press is refused, and logged, while the machine is off, an axis is unhomed, the interpreter is busy or program protect is on.
- The all-axes block lifts Z before it moves X and Y. A single `G0 X0 Y0 Z0` would drag the tool through the part.

6.8 The Manual page

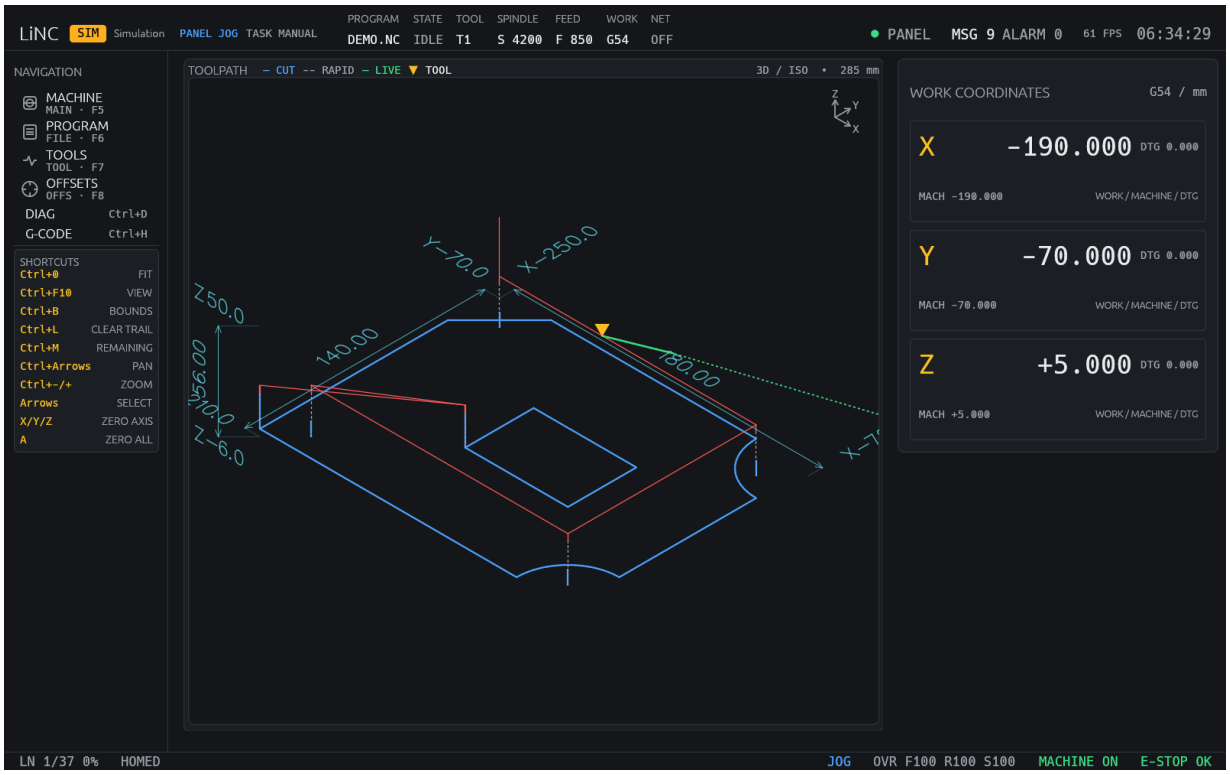


Figure 6.4: The Manual page in JOG mode: toolpath with the loaded program as a muted reference, and the work coordinates.

The Manual page is read-only apart from the increment row and the touch-off keys. It shows:

- the toolpath, with the loaded program drawn as a reference and the live trail in its own colour;
- the work coordinates, the machine coordinates and the distance to go for every axis;
- the active axis, highlighted from the panel's axis selection. An unknown selection is shown as a dash, never guessed.

TIP

Ctrl + **0** re-frames the drawing on the tool where it is now. After a long jog the trail can drag the view, and a fit brings the part and the tool back together.

7 Running a Program

A program is loaded in the Files browser, checked on the Program page, and started from the panel. LiNC never starts a program by itself and never starts one when you load it.

7.1 Loading a program

- Press **F6** for Program, then **F6** again for Files. On any other page, **Ctrl** + **O** opens Files directly.
- The right-hand pane lists the program directory. It comes from the controller's `PROGRAM_PREFIX`; there is no second copy of that path in the interface.
- The left-hand pane shows removable media. **Enter** on a file there copies it into the program directory.
- **Enter** on a program in the right-hand pane loads it.

Loading always does the same four things: it closes Files, switches to Program in the **SPLIT** layout, fits the toolpath and highlights line 1. It does not start anything, and it does not change the panel's mode.

> ⚠ **CAUTION**
>
> Only files inside the program directory are loaded. The controller will not accept a path outside it, and LiNC does not offer one.

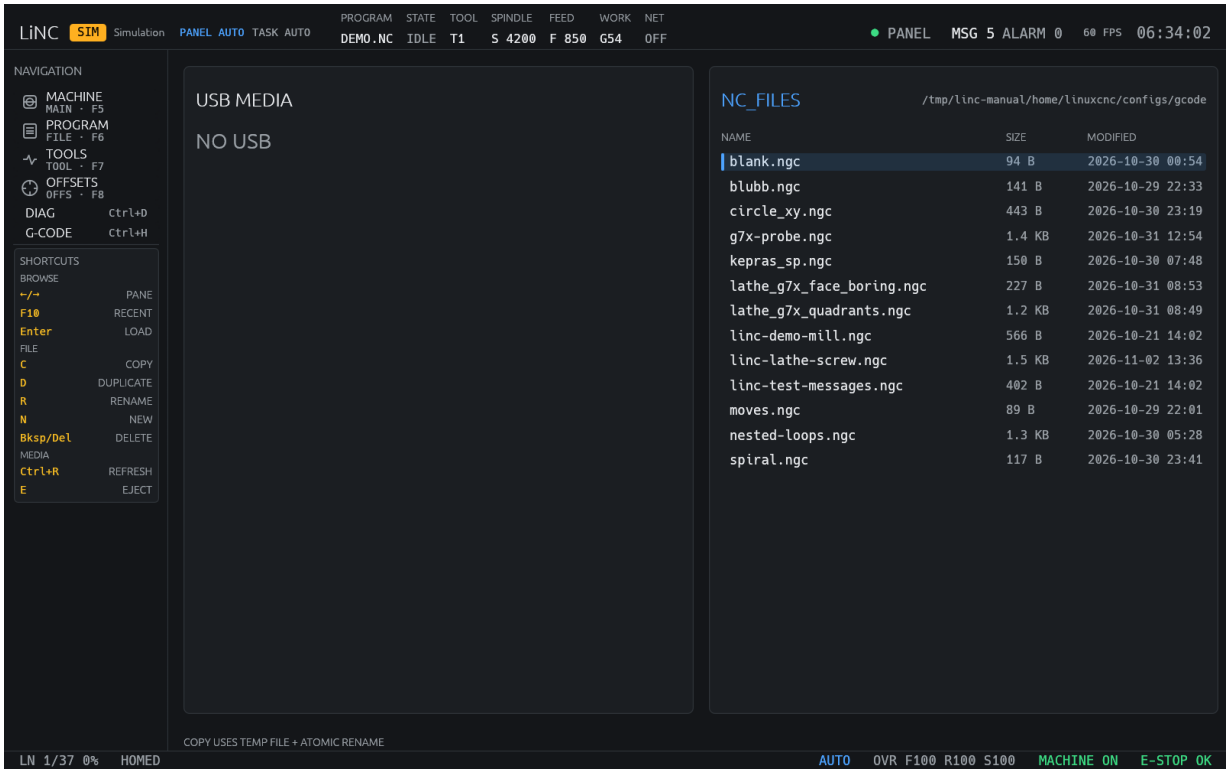


Figure 7.1: The Files browser: removable media on the left, the program directory on the right. Every action is a key; there is no button row.

7.2 Checking the program

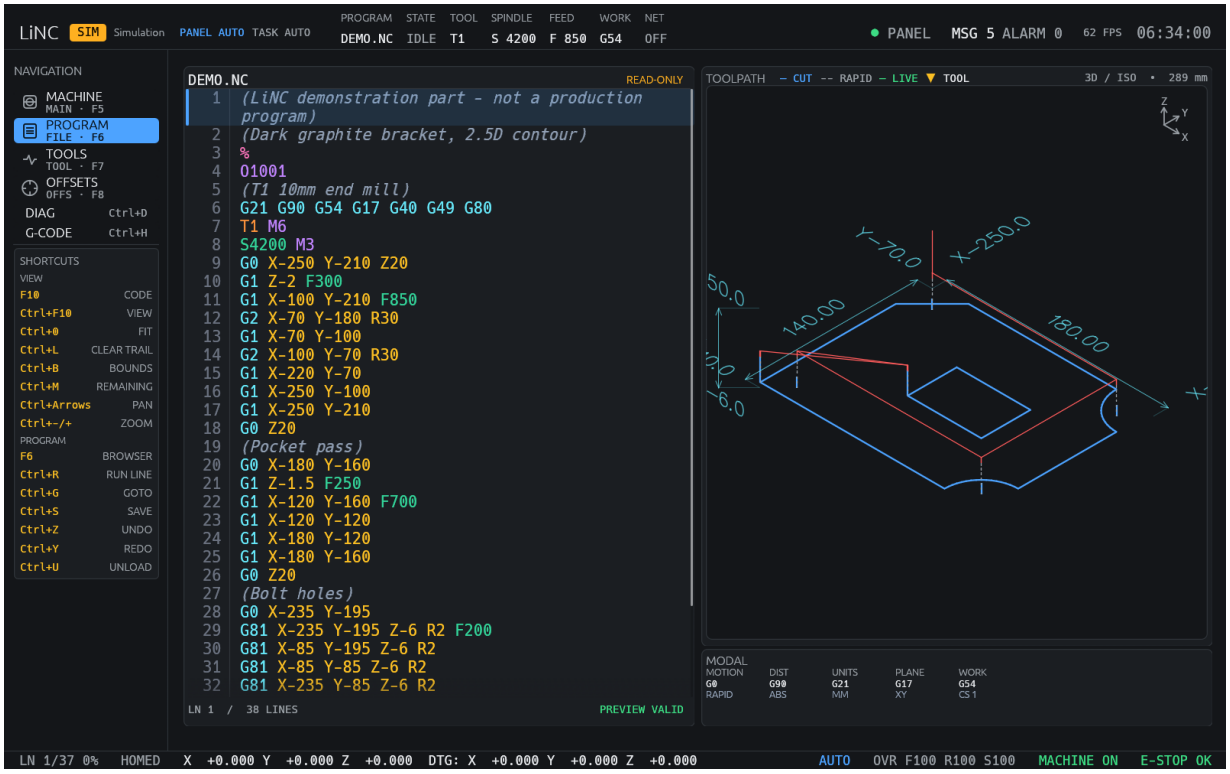


Figure 7.2: The Program page in SPLIT: editor or monitor on the left, toolpath on the right, modal codes under it.

The Program page is the checking surface, and the Machine page is the running surface.

- **The path tells you what the machine will do.** The loaded program is drawn in program colours with dotted rapids. Anything outside the machine's travel envelope is drawn solid red, split at the exact crossing, so a program that would crash is visible before it runs.
- **RED means outside the machine.** It is not a style accent and it is not a warning about the toolpath's shape.
- **The preview validates.** A parse error keeps the last valid drawing and marks the offending line; it never draws a guess.
- **The modal pane** shows the active units, plane, motion mode, work system and compensation, read from the controller as the program runs.
- **The G-code help** answers questions offline: Ctrl + H.

The Machine page carries the same toolpath plus the program card: the listing with the executing line, the modal band, the progress bar and a status word. That word reads **RUNNING**, **PAUSED** or **STOP**. It is a readout. There is no stop button on the screen.

7.3 Starting the program: cycle start

The panel's **CYCLE START** starts the loaded program. The controller gates it, and the usual conditions are:

- machine power on, no emergency stop,
- every axis homed,
- the interpreter idle: not running, not waiting for a tool change,
- the panel's mode selector in **AUTO**,
- panel data current, program protect irrelevant to running but relevant to editing.

LiNC does not issue cycle start. It shows the line and the progress, and it shows the controller's own error in the messages when a start is refused.

7.4 Starting at a line: RUN FROM LINE

Ctrl + **R** on the Program page starts the loaded program at the selected line. A line number is the one program-start input the panel cannot supply, so this is the only program start the interface issues.

- It runs only when a program is loaded, the interpreter is idle, the machine is on, no emergency stop is latched, every axis is homed, the panel data is current and the selector is in AUTO.
- A confirmation names the line and its block and states that the lines before it are skipped: work and tool offsets, feed, spindle and coolant are not restored, and motion starts from the current position.
- When a condition blocks it, the reason is shown as a message. The key does not go quiet.

7.5 Finding a line: GO TO LINE

Ctrl + **G** asks for a line number and moves the program view there. The editor, the path highlight and the Machine page's program card follow it. While the program runs the view normally follows the executing line; a line you chose with **Ctrl** + **G** or the arrow keys is kept until the next run starts.

7.6 While the program runs

- The executing line is marked in the listing with a bar and a tinted row, and the view scrolls to keep it visible.
- The bottom bar shows **`LN 12/340 34%`**: the executing line, the line count and the progress.
- The Machine page's toolpath draws the live tool trail in its own colour, separate from the program path. **Ctrl** + **L** clears the trail without touching the program.
- **Ctrl** + **M** switches to REMAINING: the trail disappears and every block already cut is eaten out of the program path, so the card shows only what is left.
- Feed hold, single block, block delete, optional stop and cycle stop are panel keys. They are wired to the controller, not to LiNC.

> **NOTE**
>
> The editor is read-only while a program runs. Clicking into the listing selects a line for the view; it does not edit the running program, because the controller runs the copy it opened, not the buffer.

7.7 Stopping and resetting

- **CYCLE STOP** on the panel stops the program and stops motion.
- **Esc** resets: it aborts the program at the controller and rewinds the interpreter. The program stays loaded and can be started again.
- A paused program is resumed by cycle start, not by RUN FROM LINE: the controller refuses a run-from-line on a paused program rather than resuming it at the wrong place.

7.8 Unloading the program

Ctrl + **U** on the Program page unloads the program and releases the external network link.

What it does, in order:

1. It refuses to run while the program is running (**`PROGRAM RUNNING — STOP FIRST`**), with unsaved changes (**`UNSAVED CHANGES — SAVE OR REVERT FIRST`**) or with nothing loaded (**`NO PROGRAM LOADED`**).
2. It sends the controller's program close, then resets the interpreter.
3. It re-reads the program name from the controller. If the controller still names a program, the unload fails loudly with **`the controller kept the program after close`** instead of pretending to succeed.

4. It shows **`UNLOAD REQUESTED — WATCH NET`**. The **`NET`** field returns to **`UP`** within a few seconds, once the radio is associated and a default route exists.

Unloading also clears the editor and the toolpath on the Program page when the controller reports that no program is loaded.

> **TIP**
>
> If the network does not come back, look at the **`NET`** field, not at the WiFi menu. **`?`** means the state is unknown; **`UP`** means the link is back. The interface writes the interlock's `net-safe` output and the machine HAL does the rest.

7.9 Program changed on disk

If the file on disk changes while it is open in the buffer, Program shows the banner **`PROGRAM CHANGED EXTERNALLY — unsaved buffer preserved. Save will use the current revision.`** The buffer does not reload itself and does not lose your edits. Decide: save over the new file, or discard and load again.

8 MDI and the G-code Editor

8.1 MDI: one block at a time

MDI is a working area, not a text box. Turn the selector to **MDI** and the Machine page opens with the MDI pane in the slot the program card normally fills.

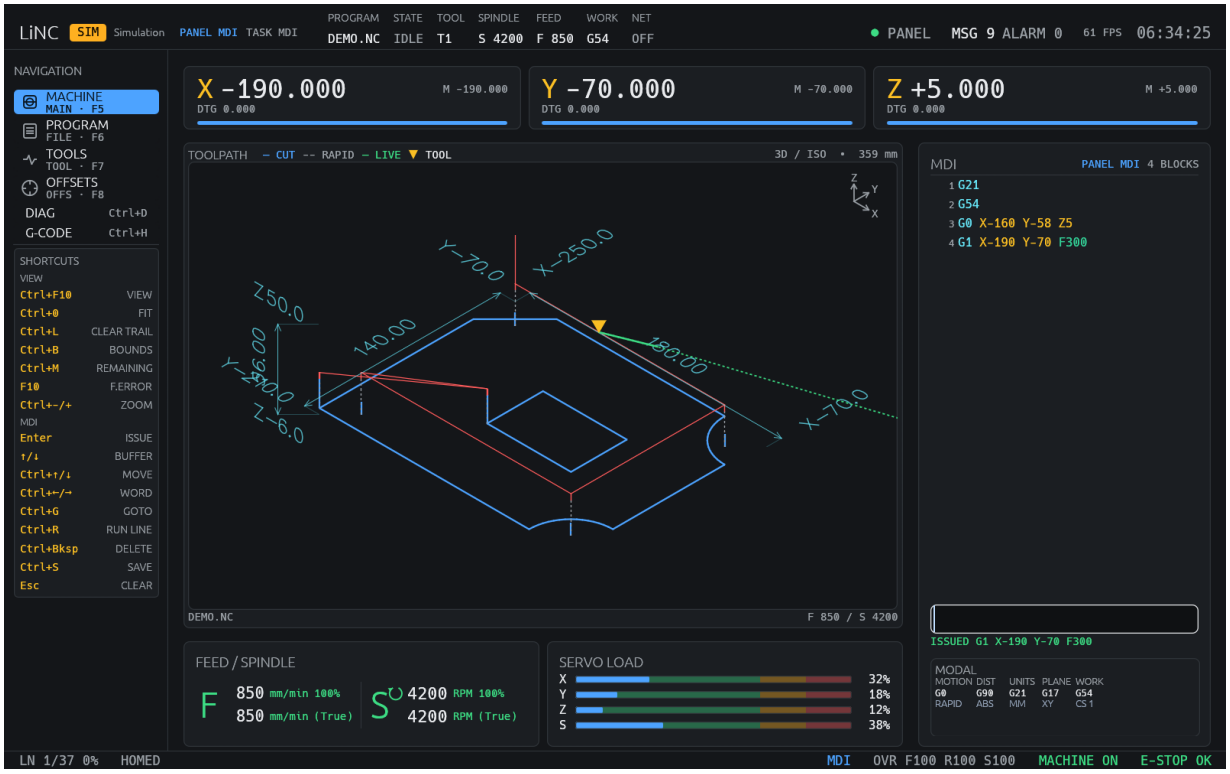


Figure 8.1: The MDI pane on the Machine page: the blocks already issued, the input line, the result strip, the modal band, and the block being typed drawn on the toolpath.

The pane, top to bottom:

- **The buffer**: every block issued in this session, in order, with a selection band on the row the arrow keys point at.
- **The input**: one syntax-highlighted line. It holds the keyboard by default, so a block can be typed as soon as the pane appears.
- **The result strip**: the last issue result, or the validation error that stopped it.
- **The modal band**: the same active codes the AUTO program card shows.

Keys

Keys	Action
Enter	Issue the block. It is committed to the buffer on success.
↑ / ↓	Move through the buffer and recall that line into the input for editing. Recall edits in place: the line stays in the listing while it is being edited.
Ctrl + ↑ / Ctrl + ↓	Move the selected buffer line up or down.
Ctrl + ← / Ctrl + →	Move by word in the input.
Ctrl + G	Go to a buffer line.
Ctrl + R	Issue the selected buffer line directly, with no confirmation.
Ctrl + Backspace	Delete the selected buffer line; with no line, clear the input.
Ctrl + S	Save the buffer as a program file.
Esc	Clear the uncommitted input. An already issued block is not cancelled.

What the pane refuses

- A block issues only when the machine is on, every axis is homed, the interpreter is idle, no program is loaded and the selector is in MDI.
- A syntax error issues nothing and marks the input.
- The controller is switched to MDI for the block and put back afterwards. No physical cycle start is involved.
- The preview of the block is refused, not clipped, when it would leave the machine's travel envelope.

The preview

The block you are typing is drawn on the Machine page's toolpath, over the loaded program, in the MDI preview colour: dashed for a rapid, solid for a cut. Its move starts at the machine's current position. This is what makes MDI safe to use for a quick approach: the move is on the screen before it is sent.

The preview understands G91: a relative move is drawn where it will go, not where its words would put it in absolute mode.

> **NOTE**
>
> An empty, unparsable or over-long block draws nothing extra. The strip says why, and the pane will refuse it.

Saving the buffer

Ctrl + S writes the buffer to a new `.ngc` file in the program directory. The dialog asks for a name and offers an **add pre/postamble** checkbox. Ticked, the file gets the profile's preamble above the buffer and its postamble below.

The buffer is kept across page changes and restarts. It is an interface working list, not a machine log.

8.2 The G-code editor

Turn the selector to **EDIT**. The Program page opens in the **CODE** layout with the editor focused and editable. The panel's selector is the authority: outside EDIT the buffer is read-only, whatever page is in front.

- **Layout.** F10 toggles SPLIT and CODE. SPLIT is for reading a program beside its path; CODE is for editing, with a narrow path thumbnail.
- **Highlighting.** G-codes, M-codes, axis letters, numbers, feed and speed, tool words, parameters and comments each have their own colour. Comments are italic.
- **Line numbers, current line and the dirty marker.** An edited buffer marks itself in the header and accents **SAVE** until it is saved.
- **The preview** updates from a debounced parse. Invalid G-code keeps the last valid drawing and marks the failing line. The preview never moves the machine.

Keys	Action
Ctrl + S	Save.
Ctrl + Z / Ctrl + Y	Undo / redo.
Ctrl + A / C / X / V	Select all, copy, cut, paste.
Ctrl + G	Go to a line.
Ctrl + D	Duplicate the current line.
Ctrl + K	Delete the current line.
Ctrl + ↑ / Ctrl + ↓	Move the current line up or down.
Ctrl + ← / Ctrl + →	Move by word.
Home / End	Start / end of the line.

Keys	Action
PgUp / PgDn	A screenful up or down, or to the start and end of a program shorter than one screen.

The editor applies the same spacing rule as MDI: a word typed against anything that is not a space gets one space in front of it, so `G1X10` arrives as `G1 X10` and `X10G0` as `X10 G0`. Nothing is inserted inside a comment or an expression, and nothing at all when text is pasted.

Saving

Saving is explicit. There is no autosave. A save is allowed only when:

- the panel is in **EDIT**,
- the interpreter is idle,
- program protect is off.

A buffer that has no file yet asks for a name. The dialog shows the program directory, appends the configured G-code extension when the name has none, refuses a name that already exists, and keeps the buffer on that path for later saves.

After writing, the controller is given the saved file again, so AUTO runs the text you saved, not the copy the interpreter opened before the edit.

> **CAUTION**
>
> Protect and a running program both refuse a save. The unsaved-changes dialog greys **SAVE** and names the blocking condition instead of letting the key do nothing.

Unsaved changes

The buffer is protected against being lost by accident:

- Leaving Program, opening another file or quitting with a dirty buffer raises a confirmation.
- **DISCARD** really drops the buffer: it is reloaded from the loaded program, or emptied when nothing is loaded, before the destination is entered.
- Leaving EDIT with a dirty buffer warns that AUTO can only start the loaded file, and offers **DISCARD BUFFER** or **KEEP**.
- Switching protect on keeps the buffer and its dirty marker, read-only, until the switch is off.

While a program runs the editor is read-only and scrolls to the executing line. The controller runs its own copy; editing the buffer mid-run was never going to change the cut.

9 Tools, Offsets and Tool Changes

9.1 The tool table

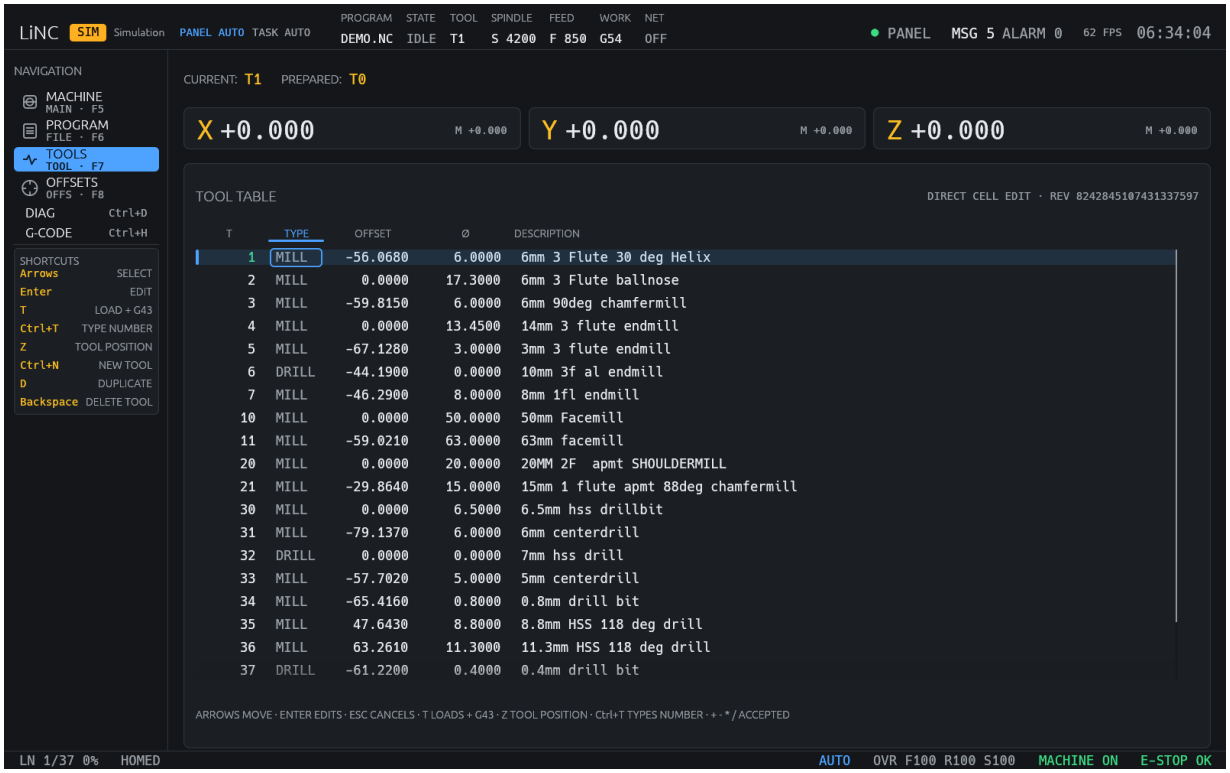


Figure 9.1: The Tools page. The table is the page: there is no separate change pane and no changer pane.

The Tools page reads and writes the controller's standard tool table, the file named by `[EMCIO] TOOL_TABLE` in the machine INI. The columns come from the table's own `TOOL_TABLE_COLUMNS`, so the page shows the schema the machine actually has.

Above the table, the current and prepared tools are named: **CURRENT: T3**, **PREPARED: T7**. A pocket column appears only on a machine with an automatic changer; a manual-change machine hides it.

Keys	Action
Arrows	Move through the table and its rows. The table takes the keyboard when the page opens.
Enter	Edit the selected cell. The whole value is selected. Enter writes it; Esc restores the previous value.
T	Load the selected row's tool: `Tn M6` and `G43 Hn` in one command, so length compensation is in effect when the change completes.
Ctrl + T	Type a tool number instead of cursoring to its row. `0` unloads the spindle.
Z	Open the tool position dialog for the tool in the spindle.
Ctrl + N	Add a row with the next free tool number.
D	Duplicate the selected row under the next free number.
Backspace	Delete the selected row after a confirmation.
S	Run one Loobric sync, when the machine has a Loobric server.

- Writing a cell reloads the table into the controller and re-applies `G43` when the edited row is the tool in the spindle. The interface waits until the controller publishes the new row, so compensation cannot run against the table it is replacing.
- The probe tool and the tool in the spindle cannot be deleted, and the probe tool number is skipped when a new row is created.
- The 'Z' column and the length field are one offset under two names: either cell writes both.
- An edited cell turns red while it holds an uncommitted value.

9.2 The tool position dialog

Z opens one line: SET <axis> POSITION, the entry, and the offset the confirm would write. It asks where the tool is now, seeded with the work coordinate the controller reports, and writes the tool offset so the work coordinate reads the entered value.

- X, Y and Z choose which coordinate is being entered.
- A Z entry updates the tool's length. X and Y update the geometry offsets the same way.
- The entry accepts arithmetic: `-12.5` and `10+2.5` both work.
- A refused write (machine off, running, protect, panel offline) keeps the dialog open and names the reason.

9.3 Tool numbers without cursoring

Ctrl + **T** opens a one-line tool-number dialog, prefilled with the tool in the spindle and fully selected, so **Enter** alone re-loads that tool.

Only a number the table carries, or `0`, is accepted. A refusal names the problem (`T7 is not in the tool table`, an empty entry, a non-numeric one) and keeps the dialog open.

9.4 Manual tool changes

On a machine with a manual changer, `M6` pauses for the change and a prompt appears – on every page, not only on Tools:

INSTALL	The tool to put in the spindle.
REMOVE	The tool to take out, with its description.
CONFIRM	The only action. It confirms that the tool is in.

The prompt follows the controller's request and cannot be dismissed. Confirming is a handshake, not motion: it is allowed while the program is running or paused, and it does not need cycle start. It raises the declared `linc.tool-change-confirm` output and holds it until the controller drops its own request, so a short pulse cannot be missed.

Until it is confirmed, the bottom bar shows **`TC T7 PENDING`**.

> **SAFETY**
>
> Check that the tool is seated and clamped before confirming. The controller continues the program immediately after the confirmation.

9.5 The tool in the spindle survives a restart

The controller starts a fresh session with an empty spindle, even when the previous session left a tool in it. LiNC remembers the tool number in a state file beside its preferences and restores it with `M61 Q`*n* and `G43 H`*n* once the machine is on, homed, idle and unprotected, and the tool is in the table.

A controller that already reports a tool wins; the remembered number is only used when the controller reports an empty spindle.

9.6 Loobric tool-table sync (optional)

A machine with a Loobric server can keep its tool table in step with it. When the profile has a `[loobric]` section, the Tools page shows the sync state in its header and accepts **S** for a manual sync.

State	Meaning
`syncing...`	A sync is running.
`tool write waiting for idle`	A local tool change is waiting to be sent.
`server changes waiting for idle`	The server has changes to bring in.
`not synced yet`	Nothing has been exchanged in this session.

Syncs wait for an idle machine. A tool write made just before starting a program is sent when the machine is idle again, not in the middle of a cut. **S** refuses with **`Loobric sync waits for idle`** while a program runs, which is the same rule the automatic sync keeps.

9.7 Work offsets

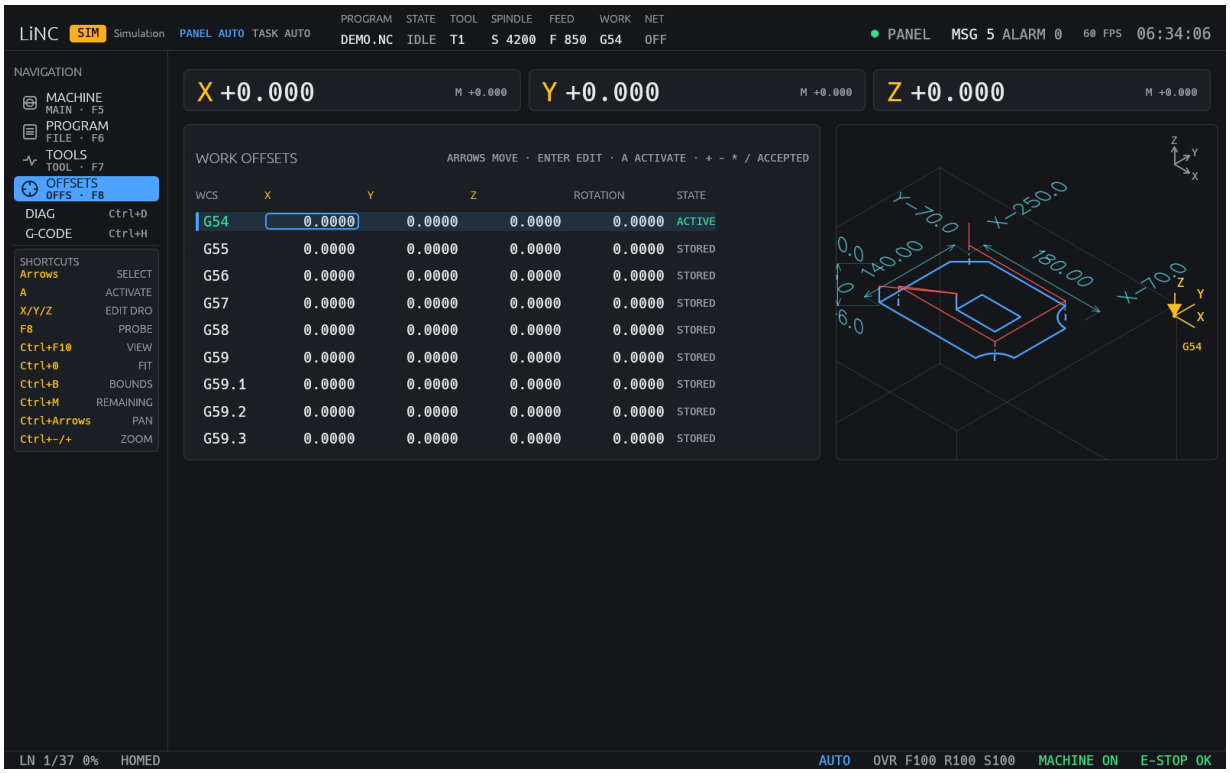


Figure 9.2: The Offsets page: the DRO row, the nine work coordinate systems with rotation, and the preview drawn in machine space.

The Offsets page holds the nine work coordinate systems the controller numbers, `G54` to `G59.3`, each with a rotation column, plus the tool offsets.

- **The DRO row** shows the active system's work coordinates large, with the machine coordinate small underneath. A work value is edited in place and accepts arithmetic; committing writes the position with `G10 L20`, so the DRO reads the value you entered and the offset moves to match.
- **A** makes the highlighted row the active system without leaving the page. The controller selects it with an MDI block, and the interface verifies the active index it reports afterwards.
- **X**, **Y** and **Z** open the DRO for that axis.
- **F8** moves on to the probe tabs; **Esc** comes back here.
- **The preview** is drawn in machine space in the work system the program selects, so the part sits where it will be cut. The view keys work as everywhere else (**Ctrl**+**F10**, **Ctrl**+**0**, **Ctrl**+**B**, **Ctrl**+**M**, **Ctrl**+arrows, **Ctrl**++/−).

G92 is deliberately not exposed here. It is a transient shift that silently moves every work system, and a field for it invited a machine state that neither the DRO nor a program could explain.

> ⚠ **CAUTION**
>
> A work offset moves the program, not the machine. The preview shows the effect before you run anything: check where the part sits after an edit, especially after a rotation.

10 Touch Probing

The probe page turns the touch probe into a guided measuring tool: choose a category, pick an illustrated operation, set the parameters, run it, read the result. The probing math is not reimplemented – the Probe Basic G-code routines ship with LiNC and run as they are.

Probing is available on a mill profile with a probe configured. On the Offsets page, **F8** moves into the probe tabs.

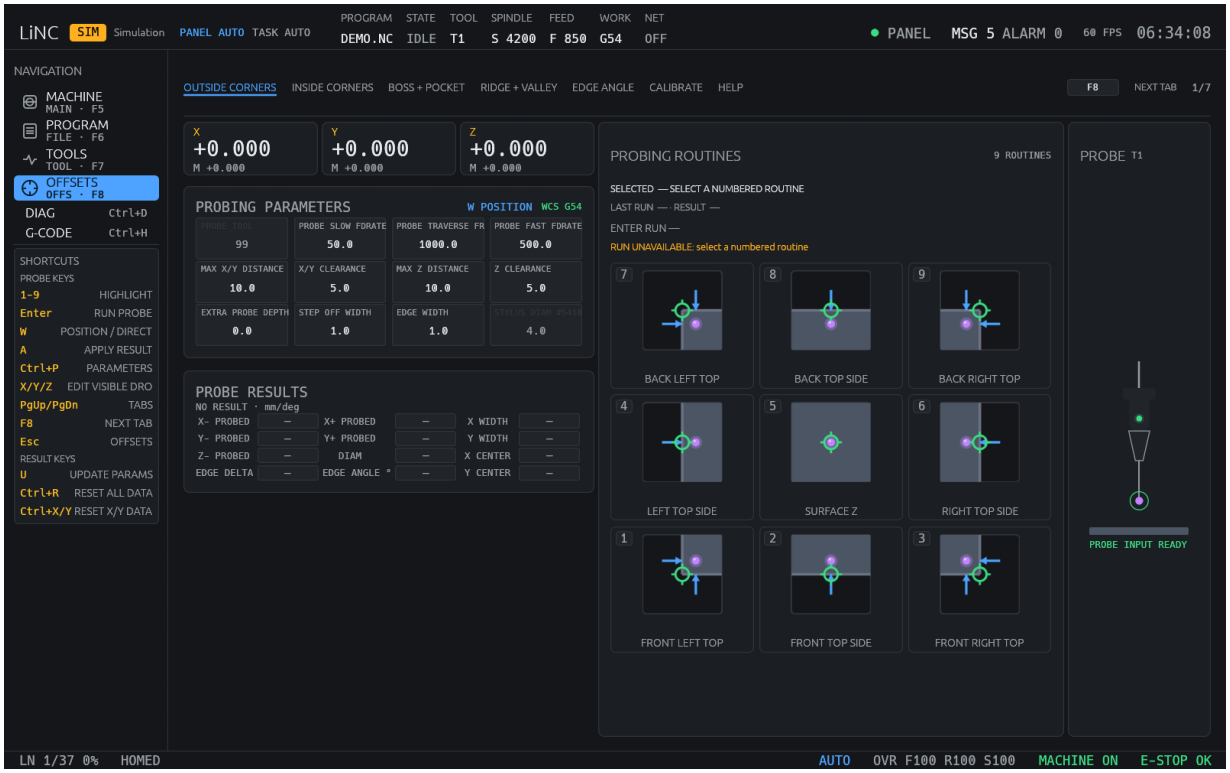


Figure 10.1: The probe page: editable DRO and parameters on the left, the routine matrix in the centre, the probe and results on the right.

10.1 The tabs

Tab	What it measures
OUTSIDE CORNERS	X/Y faces and corners from outside the part
INSIDE CORNERS	Faces and corners from inside a pocket
BOSS AND POCKET	Round and rectangular bosses and pockets: width, centre, diameter
RIDGE AND VALLEY	Ridges, valleys and their centres
EDGE ANGLE	The angle of an edge, and applying it as a work rotation
CALIBRATE	Probe calibration: round and square boss and pocket, offsets, reset
PROBE HELP	The Probe Basic help text, page by page

F8 moves to the next tab and **PgUp** / **PgDn** cycle them; **Esc** returns to Offsets. The tabs are indicators, not buttons, and take no keyboard focus.

10.2 The page

- **Top left:** an editable work-coordinate DRO. **X**, **Y** and **Z** open that axis, **Enter** writes the work position with `G10 L20`, **Esc** cancels. Work offsets themselves stay on the Offsets page; the header names the active system.
- **Below it:** the parameter card, with the probe tool, the fast, slow and traverse feed rates, the maximum X/Y and Z distance, the clearances, the extra depth, the edge width and the diameter hints. Arithmetic is accepted, as in the offset cells.
- **Centre:** the illustrated routines of the selected tab. **1** to **9** highlight a routine. **Enter** runs it.
- **Right:** the probe symbol and the results of the last run.

No routine is selected when the page opens or a tab changes. A number key is the only way to select one; **Enter** before that reports that a routine must be selected.

> **TIP**
>
> The keys that matter are listed in the pane under the navigation column: **PROBE KEYS** and **RESULT KEYS**. After an hour away from the machine, that pane is faster than this chapter.

10.3 Position and direct: the W key

W switches what a run does with its result, and the screen says which mode is active:

POSITION The default. The routine measures and records. Nothing is written to a work offset; **A** applies the last result when you decide to.

DIRECT WCO The routine writes the active work offset itself, in the issued block. It needs a valid active offset, and it is refused when protect is on.

10.4 Running a routine

Enter builds the selected `o<probe_...> call` and its argument values and issues it immediately. The controller is switched to MDI for the block and the previous task mode is restored when the interpreter returns idle. No physical cycle start is involved.

A run requires:

- machine power on and every axis homed,
- the probe tool in the spindle,
- an idle interpreter and no active program,
- valid parameters.

A validation error issues nothing and marks the failing condition.

> ⚠ **CAUTION**
>
> A probe cycle moves the machine. Check the clearance values and the probe tool before **Enter**. The fast feed rate is the approach; the slow feed rate is the contact.

10.5 The results

Results come from the subroutine outputs and are shown as named slots: **X WIDTH**, **X CENTRE**, **Y WIDTH**, **Y CENTRE**, **X− FACE**, **X+ FACE**, **Y− FACE**, **Y+ FACE**, **Z SURFACE**, **AVG DIAM** and **EDGE DELTA**.

Keys	Action
A	Apply the last result to the active work offset. Protect must be off.
U	Update the probe parameters from the last result.
Ctrl + **R**	Reset all probe data.
Ctrl + **X** / **Ctrl** + **Y**	Reset the X or the Y data.

The results panel carries no buttons: those keys are the whole surface.

10.6 Edge angle and work rotation

On **EDGE ANGLE**, R applies the measured angle as the active work coordinate rotation. Check the Offsets page afterwards: the rotation appears there, and the preview shows the part rotated around its zero.

10.7 Calibration

The **CALIBRATE** tab measures the probe against a known feature and stores the calibration offset. Its fields – known width or diameter, calibration axis and the calibration offset itself – live on the same panel.

- 1 to 4 highlight a calibration routine; Enter runs it.
- V switches the calibration between the averaged XY result and the single X or Y result.
- Ctrl + P reaches the calibration dimensions.
- Ctrl + R resets the calibration through the Probe Basic reset routine.

10.8 When the probe is hit

If the probe input rises during a move that is not a probe, the controller aborts the move and reports an operator error. LiNC latches it: a red banner carries the controller's text on every page, motion and MDI are blocked, and the machine is switched off unless it was paused with nothing moving.

Esc clears the latched trip from any page. Re-powering afterwards needs a fresh ON at the panel. The full behaviour is in chapter 2.

> **NOTE**
>
> The probe input is also shown in the bottom bar, but only while the probe tool is in the spindle: **PROBE READY** or **PROBE TRIPPED**. There is no probe field for a cutter in the spindle, because there is no probe to read.

11 Diagnostics and Troubleshooting

Ctrl + D opens Diagnostics. The page is read-only: nothing here jogs, cycles, resets or acknowledges. It exists to answer three questions – is the controller healthy, what did it say, and what is the machine actually doing – and to let you read the answer without a terminal.

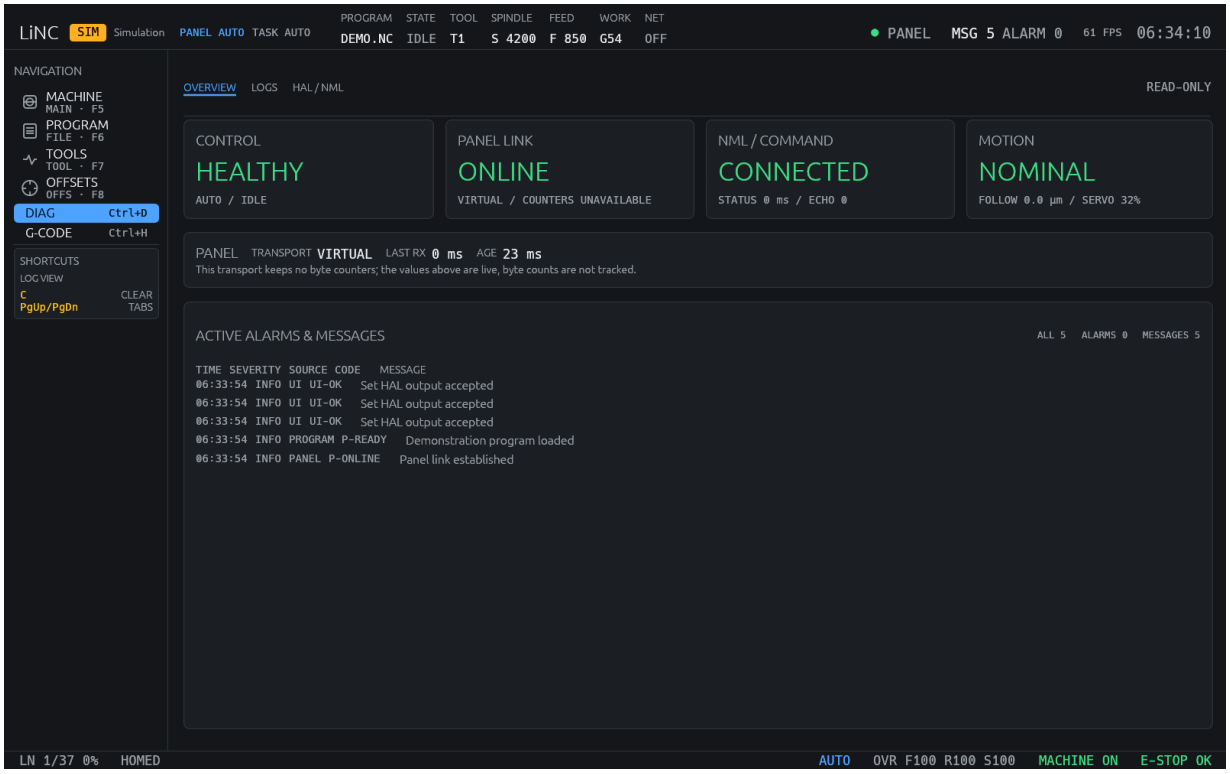


Figure 11.1: Diagnostics, OVERVIEW tab: four health cards, the panel link strip, and the active alarms and messages.

11.1 OVERVIEW

Card	Values	Detail line
CONTROL	**HEALTHY** or **E-STOP**	task mode / interpreter state
PANEL LINK	**ONLINE** or **OFFLINE**	transport, port, CRC errors, timeouts
NML / COMMAND	**CONNECTED** or **LOST**	status age, echo serial
MOTION	**NOMINAL** or **ELEVATED**	largest following error in µm, largest servo load in percent

ELEVATED means a following error above 50 µm. It is a warning threshold, not a fault: look at the axes and the load, and at the machine's own limits.

Below the cards, the **panel link strip** shows the transport, the last receive time and its age. A transport that keeps no byte counters says so instead of showing invented ones.

11.2 LOGS

The event history, newest entries first, with the time, severity, source, code and message. The severity and source filters are toggles; the status line reports which are on, and shows **ALL** for a group with nothing selected.

Keys	Action
/	Focus the search line. Matches message, code and source.
Enter	Leave the search line and take the list.
Esc	Leave the list and return to the search line.
I W E C	Toggle INFO, WARN, ERROR and CRIT severity filters.
P G B T H	Toggle PANEL, PROGRAM, PROBE, TOOL and HAL sources.
N M O U Y	Toggle NML, MOTION, I/O, UI and SYSTEM sources.
A	Clear all filters.
PgUp / PgDn	Scroll a list that overflows, otherwise change tab.

> **NOTE**
>
> The letter toggles act only while the list holds the keyboard. In the search line the same letters are text – which is how you search for a word containing an e.

11.3 HAL / NML

The **NML / COMMAND** status line at the top, then the HAL browser: pins grouped by component, each with its type, direction, value and signal, and a separate signal list.

- **/** focuses the search, **Enter** takes the list, **Left** and **Right** switch between pins and signals, and **Esc** returns to the search.
- **F** and **S** refresh faster or slower. The header shows the age of the rows it drew.
- The browser shells out to `halcmd show pin` and `halcmd show sig` on a background thread. Without a running controller or without `halcmd` it says so and shows nothing; it never fills the tree in from the machine snapshot.
- Nothing on this tab writes to HAL.

> **TIP**
>
> The HAL browser is the fastest way to see whether a panel key or a limit switch actually reaches the controller: search for the pin, press the key, watch the value change.

11.4 Messages

M opens the full-screen message list from any page. It carries the same rows as the LOGS tab, without filters, and closes with **Enter** or **Esc**.

11.5 Where the session log goes

The interface writes its own log (errors, warnings, the start-up line naming the profile and the loaded program) to standard error. The machine's launcher redirects it into the profile's log directory, for example `logs/linc.log`. Read it when the interface closed with a message but the screen is gone.

11.6 Troubleshooting

Symptom	Cause	Action
`PANEL OFFLINE` banner	Panel service not running, or the serial link is down	Start the panel service; check the cable. The emergency stop still works.

Symptom	Cause	Action
PANEL MODE INVALID	Selector between detents, or a wiring fault	Turn the selector to a definite position. LiNC refuses mode-dependent actions until it is valid.
MACHINE OFF blinks	Machine power is off	Press **ON**. The blink marks a fault, not an idle machine.
NOT HOMED	No reference	Home the axes. Programs and MDI refuse to run.
LIMIT X	An axis is on a limit switch	Jog off the switch; use **Ctrl** + **Shift** + **O** if the switch blocks the jog, then switch the override off.
PROTECT ON	The key switch is on	Turn it off. Writing, importing and applying are blocked while it is on.
TC T7 PENDING	A manual tool change is waiting	Install the tool and confirm the prompt.
Save refused	Not in EDIT, program running, or protect on	The unsaved-changes dialog names the blocking condition.
NETWORK BUSY	The external link is up while a program is loaded	Wait for the interlock to settle, or unload with **Ctrl** + **U** if you are finished.
NET stays **UP** with a program loaded	The interlock is disabled or set to `keep_up`	Check `[network]` in the profile.
Remote access is dead	A program is loaded, so the link is down	That is the design. Unload at the machine, or use a profile without the interlock.
Unload refused	Program running, buffer dirty, or nothing loaded	Stop first, save or revert, or check the program name.
the controller kept the program after close	The controller did not release the program	Stop the session at the panel and restart it; report the message with the log.
PROBE TRIPPED	The probe input rose during a non-probe move	**Esc** clears the latch. Find out what hit the probe before powering on again.

Symptom	Cause	Action
? in a status field	The data is stale	The interface fails closed on stale data. Wait for the next poll; if it stays, the controller connection is the problem.
Frame rate below 30 FPS	The interface is doing too much	Look at Diagnostics; close the help article or the HAL browser if they are open.
HALCMD UNAVAILABLE	No `halcmd` on the machine, or no session	The browser has nothing to read. The rest of the page still works.
Loobric sync waits for idle	A program is running	Sync happens when the machine is idle.

CAUTION

A screenshot or a log is worth more than a memory. Before restarting a session after a fault, open Diagnostics and read the last error entries; they name the block, the axis or the condition.

12 Quick Reference

12.1 Keys

Keys	Action
Navigation	
F5	Machine page, or the page belonging to the panel mode
F6	Program; again for Files; in Files, back to Program
F7	Tools
F8	Offsets; again for the probe tabs, cyclically
Ctrl + D	Diagnostics; duplicate line while the editor has the keyboard
Ctrl + H	G-code help
Ctrl + F	Search inside the help article (Help page only)
Ctrl + O	Files (not on Program; F6 there)
M	Messages
Esc	Back, cancel, or reset the program
Ctrl + Q	Quit (refused while running or moving)
Toolpath view, on every page that draws one	
Ctrl + F10	Cycle ISO, X, Y, Z
Ctrl + 0	Fit
Ctrl + L	Clear the live trail
Ctrl + B	Cycle bounds
Ctrl + M	REMAINING
Ctrl +arrows	Pan
Ctrl + + / Ctrl + -	Zoom
Program and editor	
F10	SPLIT / CODE layout
Ctrl + R	Run from line
Ctrl + G	Go to line

Keys	Action
Ctrl + U	Unload the program and release the link
Ctrl + S	Save
Ctrl + Z / Ctrl + Y	Undo / redo
Ctrl + A / C / X / V	Select all, copy, cut, paste
Ctrl + K	Delete the current line
Ctrl + ↑ / ↓	Move the current line
Ctrl + ← / →	Move by word
MDI pane	
Enter	Issue the block
↑ / ↓	Recall a buffer line
Ctrl + ↑ / ↓	Move the selected buffer line
Ctrl + G	Go to a buffer line
Ctrl + R	Issue the selected buffer line
Ctrl + Backspace	Delete the selected buffer line
Ctrl + S	Save the buffer as a program
Esc	Clear the input
Manual and homing pages	
1 … 9	Select the INC step (INC only)
X / Y / Z	Zero that work axis
A	Zero every axis
O	Toggle the limit-switch override
Tools	
T	Load the selected tool with G43
Ctrl + T	Type a tool number
Z	Tool position dialog
Ctrl + N	New tool
D	Duplicate tool
Backspace	Delete tool
S	Loobric sync (when configured)
Files	
Enter	Open a folder, copy from USB, load a program
C / D / R / N	Copy, duplicate, rename, new

Keys	Action
Backspace	Delete (also Delete)
E	Eject
Ctrl + R	Refresh
Probe	
1 … 9	Select a routine
Enter	Run the routine
W	Position / direct work-offset mode
A	Apply the last result
U	Update the parameters
Ctrl + P	Edit the parameters
Ctrl + R	Reset the probe data
Ctrl + X / Y	Reset X / Y data
R	Apply the edge angle as a rotation
Diagnostics	
/	Search
I W E C	Severity filters
P G B T H	Panel, program, probe, tool, HAL filters
N M O U Y	NML, motion, I/O, UI, system filters
A	Clear the filters
C	Clear the log view (OVERVIEW)
F / S	HAL browser: refresh faster / slower

12.2 One key, several meanings

These keys are page-local. The shortcut pane always shows the meaning that applies now.

Key	Where it means something else
A	Offsets: activate the row. Probe: apply the result. Manual/homing: zero all axes. Diagnostics: clear the log filters.
O	Manual/homing: limits override. Elsewhere: Ctrl + O opens Files.
Ctrl + Shift + O	Limit-switch override, on every page.
T	Tools: load the selected tool.
D	Editor: duplicate line. Files: duplicate file. Tools: duplicate tool.
S	Tools: Loobric sync. Editor: save (with Ctrl).
C	Diagnostics: clear the filter or the view.
R	Program: run from line (with Ctrl). Files: refresh. Probe: apply rotation. Diagnostics: reset the view.
U	Program: unload (with Ctrl).
W	Probe: position / direct mode.
X / Y / Z	Offsets and probe: edit the DRO. Manual/homing: zero the axis. Tools: tool position.

12.3 The panel, in short

Control	Owns
ON / OFF	Machine power
EMERGENCY STOP	Latching stop; release clears, power-on is separate
MODE	EDIT, AUTO, MDI, JOG, INC, MPG, HOME, PROG.HOME
FEEDRATE / RAPID / SPINDLE OVERRIDE	Three override percentages
CYCLE START / CYCLE STOP	Program start and stop
Axis keys and RAPID	Jog, home, and program zero, by mode
Machine function keys	Spindle, coolant, chuck, lube, chip, lamp, changer, K1–K4
PROGRAM PROTECT	Locks every write

12.4 Link states

NET	Meaning
DOWN (green)	Link down, as the interlock wants while armed
UP (muted)	Link up with no program motion
UP (amber)	Link up while program motion runs: look at this one
MEAS	A link measurement is running
? (amber)	Unknown or stale; fails closed
OFF (muted)	No interlock in this profile

12.5 Status words

Word	Meaning
IDLE / **RUNNING** / **PAUSED**	Program state in the top bar
HOMED / **HOMED X** / **NOT HOMED**	Homing state in the bottom bar
TC Tn PENDING	Manual tool change waiting for confirmation
PROTECT ON	Program protect is on
PROBE READY / **PROBE TRIPPED** / **PROBE ?**	Probe input, when the probe is in the spindle
E-STOP ACTIVE / **E-STOP OK** / **E-STOP ?**	Emergency stop, or stale data
MACHINE ON / **MACHINE OFF**	Machine power; OFF blinks
LIMITS OVERRIDE ON	Blinking; switches are ignored
PANEL ONLINE / **PANEL OFFLINE**	Panel link

12.6 Where things live

Item	Location
Machine profile	chosen at start-up with `-config` or `LINC_CONFIG`
Shared defaults	`configs/base.toml`
Machine INI	named by the profile; authoritative for travel, speed, units, homing
Tool table	`[EMCIO] TOOL_TABLE`, relative to the INI
Program directory	`[DISPLAY] PROGRAM_PREFIX`, relative to the INI
Probing routines	`[RS274NGC] SUBROUTINE_PATH`
Preferences	`~/.config/linc/`, as `<profile>`
Safety-notice acceptance	in the preferences file, with the notice version
Remembered tool	`<profile>.state.toml`, beside the preferences
Session log	the profile's log directory, written by the launcher

12.7 Safety rules, in short

SAFETY

- The panel is the only place motion starts.
- Release the emergency stop, then press **ON**: two actions.
- Never leave the limits override on.
- Red path means outside the machine: find out why before running.
- Unload the program before you rely on the network link.
- Confirm a manual tool change only when the tool is seated and clamped.